

Human iPSC-derived Microglial Cells Integrated into Mouse Retina and Recapitulated Features of Endogenous Microglia

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Abstract

Microglia exhibit both maladaptive and adaptive roles in the pathogenesis of neurodegenerative diseases and have emerged as a cellular target for central nervous system (CNS) disorders, including those affecting the retina. Replacing maladaptive microglia, such as those impacted by aging or over-activation, with exogenous microglia that can enable adaptive functions has been proposed as a potential therapeutic strategy for neurodegenerative diseases. To investigate microglia replacement as an approach for retinal diseases, we first employed a protocol to efficiently generate human-induced pluripotent stem cells (hiPSC)-derived microglia in quantities sufficient for *in vivo* transplantation. These cells demonstrated expression of microglia-enriched genes and showed typical microglial functions such as LPS-induced responses and phagocytosis. We then performed xenotransplantation of these hiPSC-derived microglia into the subretinal space of adult mice whose endogenous retinal microglia have been pharmacologically depleted. Long-term analysis post-transplantation demonstrated that transplanted hiPSC-derived microglia successfully integrated into the neuroretina as ramified cells, occupying positions previously filled by the endogenous microglia and expressed microglia homeostatic markers such as P2ry12 and Tmem119. Further, these cells were found juxtaposed alongside residual endogenous murine microglia for up to eight months in the retina, indicating their ability to establish a stable homeostatic state *in vivo*. Following retinal pigment epithelial (RPE) cell injury, transplanted microglia demonstrated responses typical of endogenous microglia, including migration, proliferation, and phagocytosis. Our findings indicate the feasibility of microglial transplantation and integration in the retina and suggest that modulating microglia through replacement may be a therapeutic strategy for treating neurodegenerative retinal diseases.

eLife assessment

The authors have improved a method to differentiate human iPSC-derived microglial cells with immune responses and phagocytic abilities; and through transplantation into the adult mouse retina, the authors further demonstrated their integration and occupation of native microglial cell space, and functional response to retinal injuries. The study is **important** and the data are **convincing** for potential microglial replacement therapy to treat retinal and CNS diseases.

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Introduction

Microglia are the innate immune cells of the central nervous system (CNS), including the retina, and play pivotal roles in neuronal (Puñal et al., 2019; Anderson et al., 2019; Huang et al., 2012) and vascular development (Checchin et al., 2006; Ritter et al., 2006; Kubota et al., 2009), normal synapse formation (Stevens et al., 2007; Schafer et al., 2012; Schafer et al., 2015; Hong et al., 2016), maintaining local homeostasis in the neural environment (Li et al., 2018; Colonna et al., 2017; Sierra et al., 2013) and the regulation of immune activity (Okunuki et al., 2019). Conversely, they are also implicated in driving pathologic progression in various retinal diseases, including age-related macular degeneration (AMD) (Combadière et al., 2007; Ma et al., 2009, 2012; Karlstetter et al., 2015), glaucoma (Bosco et al., 2019; Ramírez et al., 2020), diabetic retinopathy (Altmann C et al., 2018; Xu et al., 2017), and uveitis (Broderick et al., 2002; Okunuki et al., 2019; Zhou et al., 2020).

Under homeostatic conditions in the adult retina, microglial cells are predominantly distributed in the IPL and OPL and vigilantly survey environmental changes through dynamic surveying behavior in their ramified processes (Lee et al., 2008). Their presence and homeostatic function are crucial for maintaining normal retinal functions, including the maintenance of synaptic function and integrity (Wang et al., 2016). Under normal conditions, microglial cells sustain equilibrium in their endogenous numbers via slow self-renewal (Réu et al., 2017). However, with the onset of pathology, this homeostasis can be disrupted following microglia activation, migration, and proliferation. Microglia repopulation in the retina following perturbation is achieved through both the proliferation of endogenous microglia and the infiltration of peripheral monocytes (Ma et al., 2017; Huang et al., 2018; Zhang et al., 2018).

Studies of microglia cell repopulation have indicated that retinal resident microglia can sustain their population with the local microglial cell dividing and migration if any perturbations do not exceed the threshold of the recovery speed by local neighbor microglia. However, in cases of more severe retinal injury or infection that cause significant redistribution of endogenous microglia, the reestablishment of retinal microglial homeostasis will, in addition, involve peripheral monocytes that infiltrate into the retina to take up residence as macrophages. The ability of the retina to incorporate exogenous monocytic cells suggests that microglia cell replacement employing exogenously introduced microglia may be feasible and can exert therapeutic effects post-injury. Inhibiting retinal microglia over-activation has shown efficacy in animal models of retinal injury (Zhao et al., 2011; Karlstetter et al., 2015; Au et al., 2022) and potential signal in early-phase clinical trials (Cukras et al., 2012). These observations suggest that the depletion of maladaptive microglia in pathological contexts and their replacement with microglia that have a more homeostatic phenotype may constitute a potential therapeutic strategy.

Studies of microglia have largely been performed in rodent-derived models, largely due to the accessibility of various transgenic disease models. However, several studies have indicated that genetic and functional differences exist between murine and human microglia (Dawson et al., 2018 [↗](#); Friedman et al., 2018 [↗](#); Ueda et al., 2016 [↗](#)). For instance, microglia-expressed genes CD33 and CR1, which have been associated with Alzheimer's disease (AD) risk in genome-wide association studies (GWAS), lack reliable orthologues in mice (Hasselmann et al., 2020 [↗](#)). Additionally, over half of the AD risk genes that are enriched in microglia demonstrate <70% sequence homology between humans and mice (Hasselmann et al., 2020). There are also significant differences in the levels of protein expression of some microglia-expressed complement factors and inflammatory cytokines between humans and mice (Galatro et al., 2017 [↗](#); Gosselin et al., 2017 [↗](#); Smith & Dragunow, 2014 [↗](#)). As a result, microglia from murine models may not accurately represent those found in human conditions (Burns et al., 2015 [↗](#)), limiting their translational potential.

To study microglia of human origin, some investigators have attempted to isolate microglial cells from human tissue. However, owing to limitations in sample availability, and the rapid transcriptomic changes that ex vivo microglia undergo post-isolation, these studies have been technically constrained (Bohlen et al., 2017 [↗](#); Butovsky et al., 2014 [↗](#); Gosselin et al., 2017 [↗](#)). As an alternative to primary microglia, human-induced pluripotent stem cells (iPSCs) offer a wealth of possibilities and have been increasingly used to generate human microglia via differentiation in vitro (Muffat et al., 2016 [↗](#); Pandya et al., 2017 [↗](#); Abud et al., 2017 [↗](#); Douvaras et al., 2017 [↗](#); Haenseler et al., 2017 [↗](#); Takata et al., 2017 [↗](#)). This approach has enabled the generation of a large quantity of cells of a specified genetic background, enabling the creation of in vivo human microglia cell models through xenotransplantation of hiPSC-derived microglia into the murine CNS (Abud et al., 2017 [↗](#); Svoboda et al., 2019 [↗](#); Parajuli et al., 2021 [↗](#); Xu et al., 2020 [↗](#); Chadarevian et al., 2023 [↗](#)).

In this study, we adopted a previously published protocol appropriate for culturing hiPSC-derived microglial cells (Muffat et al., 2016 [↗](#); Pandya et al., 2017 [↗](#); Abud et al., 2017 [↗](#); Douvaras et al., 2017 [↗](#); Haenseler et al., 2017 [↗](#); Takata et al., 2017 [↗](#)). We characterized microglia differentiation by examining the RNA and protein expression levels of microglia-enriched genes. We also assessed the inflammatory responses and phagocytic functions of hiPSC-derived microglia *in vitro*. We then established a human iPSC-derived microglia cell model through the xenotransplantation of human iPSC-derived microglial cells into the retina of an adult mouse. We found that when transplanted by subretinal injection, human iPSC-derived microglial cells were able to migrate into the retina where native retinal microglia reside and acquire a morphology resembling endogenous mouse microglia, and express microglia signature markers. These grafted cells persisted in the mouse retina for at least eight months and responded to RPE cell injury in ways resembling endogenous mouse microglia. Xenografting of hiPSC-derived microglia into mouse retina has the promise of being used to create in vivo models of retinal disease and injury to evaluate the preclinical efficacy of potential therapeutic agents, as well as to evaluate microglia transplantation itself as a potential therapeutic intervention.

Results

Differentiation and characterization of human iPSC-derived microglia

We used five distinct human iPSC lines for microglia cell differentiation, including the first-available iPSC line, KYOUDXR0109B, from ATCC, and four other lines (NCRM6, MS19-ES-H, ND2-AAVS1-iCAG-tdTomato, and NCRM5-AAVS1-CAG-EGFP, all from the National Heart, Lung, and Blood Institute (NHLBI), NIH). Our approach to microglia differentiation was informed by our previous work with primary mouse retinal microglia cell culture (Ma et al., 2009 [↗](#)) and a variety of

established microglia cell differentiation protocols (Muffat et al., 2016 [↗](#); Pandya et al., 2017 [↗](#); Abud et al., 2017 [↗](#); Douvaras et al., 2017 [↗](#); Haenseler et al., 2017 [↗](#); Takata et al., 2017 [↗](#)). We opted for the myeloid progenitor/microglia cell floating culture method (Van et al., 2013; Haenseler et al., 2017 [↗](#)) for its simplicity, efficiency, and consistency, which enables the generation of a large and uniform population of microglial cells.

The differentiation process involved three key stages: embryoid body formation, myeloid progenitor cell generation, and microglia cell maturation (**Fig. 1 A-D** [↗](#)). Following myeloid differentiation, floating myeloid progenitor cells were harvested and allowed to differentiate further for two weeks in 6-well plates under conditions promoting microglial differentiation. We modified this step to include additional differentiation factors in the differentiation medium, including IL34, CSF1, CX3CL1, TGFB1, and TGFB2. We found that these factors promoted microglial morphological ramification and process elongation. Immunohistochemical analysis of the resulting cells showed that among CD34(+) cells, 98.6% were immunopositive for IBA1 (**Fig. 1E, F** [↗](#)), and 98.5% were immunopositive for P2RY12 (**Fig. 1E, G** [↗](#)). Immunostaining with myeloid cell markers CX3CR1, CD68, and CD11b showed positivity in 88%, 99.7%, and 94.3% respectively, demonstrating the high efficiency of differentiation achieved by this procedure (**Fig. 1. Suppl** [↗](#)). Most of the resulting hiPSC-derived microglia showed spindle-shaped morphologies, with some displaying short ramifications in their processes (**Fig. 1E** [↗](#)), resembling those observed in primary mouse retinal microglia cultures (Ma et al., 2009 [↗](#)). Floating myeloid progenitor cell harvest could be performed repeatedly over three months following culture establishment, providing a steady and consistent supply for further microglia differentiation and generation.

A comparative RNAseq analysis between differentiated hiPSC-derived microglia collected at the end of the protocol vs. floating myeloid progenitor cells revealed a significant upregulation of microglia-enriched gene expression following microglial differentiation, including *CX3CR1*, *P2RY12*, *P2RY13*, *AIF1*, *TREM2*, *GPR34*, *CD53*, *CTSS*, *C3AR1* (**Fig. 2A-C** [↗](#)). Moreover, hiPSC-derived microglia exhibited higher expression of genes associated with inflammation, apoptosis regulation, phagocytosis, lipid metabolism, and immune responses. The floating myeloid progenitor cells showed higher expression of hematopoietic/myeloid cell lineage genes (**Fig. 2D** [↗](#)). Ingenuity Pathway Analysis (IPA) of differentially expressed genes identified IL6, IL1 β , and STAT3 as central signaling hubs critical to the regulation of inflammatory responses in microglia (**Fig. 2E** [↗](#), **Fig. 2 Suppl A** [↗](#), Suppl Table 3). We also compared the gene expression profiles between differentiated hiPSC-derived microglial cells vs. human microglia isolated ex vivo from the fetal and adult brain (**Fig. 2. Suppl. B, C** [↗](#) and D; Suppl. table 1 and table 2) (Abud et al., 2017 [↗](#); Douvaras et al., 2017 [↗](#); Muffat et al., 2016 [↗](#); Böttcher et al., 2019 [↗](#); van der Poel et al., 2019 [↗](#); Eisenberg et al. 2013). The results of the correlation analysis indicated that hiPSC-derived microglial cells demonstrated an expression profile comparable to those in fetal and adult microglia in vivo but were more distinct from those in monocytes and inflammatory monocytes. Thus, our method of obtaining differentiated microglia can be employed to reliably and efficiently generate a large population of homogenous functional microglial cells of human origin.

Human iPSC-derived microglia show inflammation responses and phagocytosis activity

Microglia play crucial roles in mediating inflammatory responses to stimuli and in phagocytosing pathogens. To investigate these functions further, we stimulated hiPSC-derived microglia with lipopolysaccharide (LPS) and analyzed their responses. Transcriptomic profiling using bulk RNAseq revealed that the primary responses to LPS stimulation involved changes in *IL6*, *IL1 β* , *IL1 α* , *TNFA*, and *IFN γ* signaling (**Fig. 3A** [↗](#), Suppl. Table 3), indicating the ability of hiPSC-derived microglia to demonstrate classical activation. This was confirmed through expression analysis using QRT-PCR and protein multiplex profiling, which showed a 50- to 800-fold increase in the

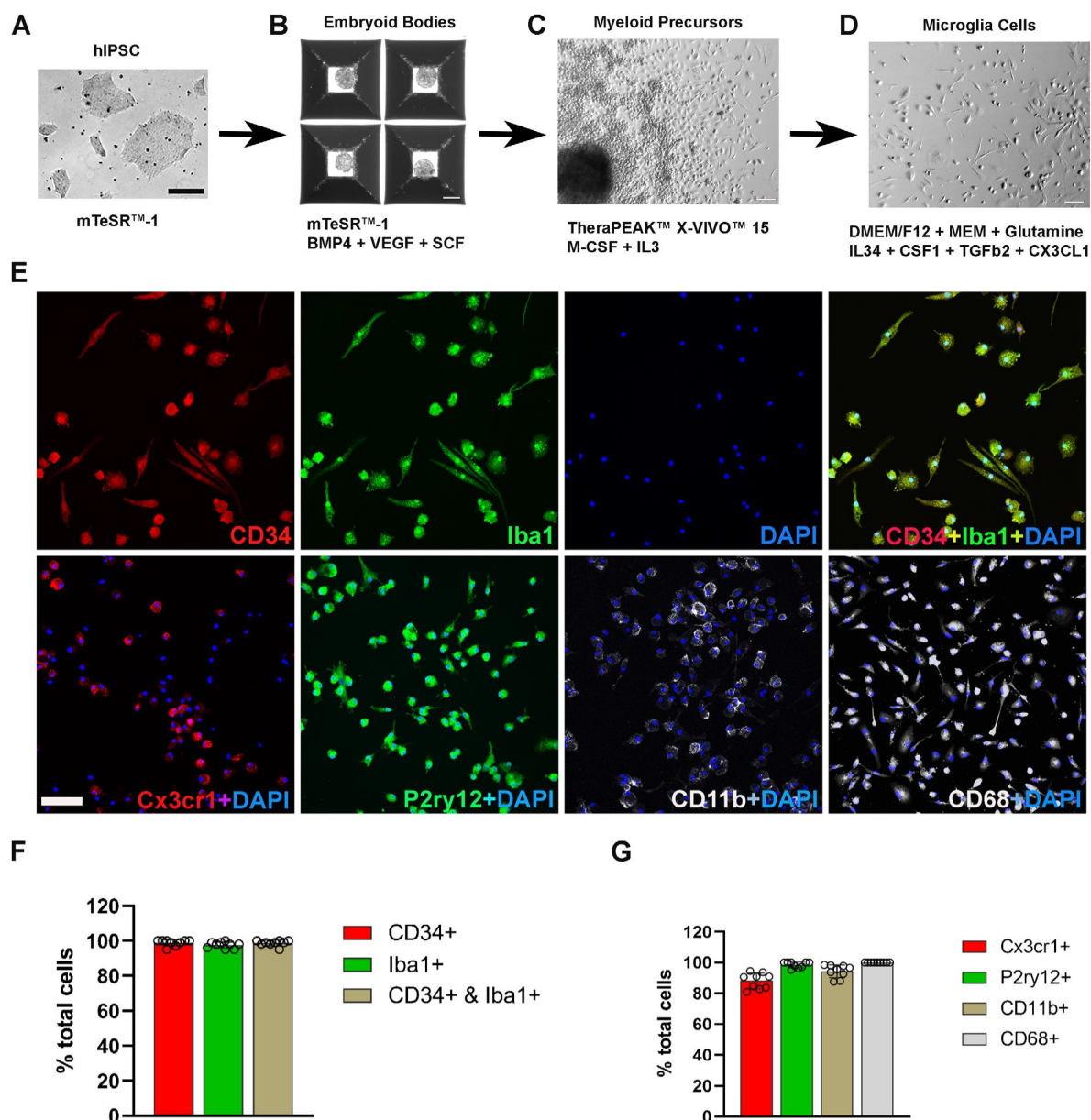


Figure 1.

Differentiation and characterization of human iPSC-derived microglia.

(A) Human iPSCs were cultured in a 6-well plate. Scale bar = 200µm. (B) Embryoid bodies formation was enabled in AggreWell™800 plate at day eight in culture medium mTeSR1 plus BMP4, VEGF, and SCF. Scale bar = 200µm. (C) Image of a myeloid precursor cluster following 1-month culture of embryoid bodies in TheraPEAK™ X-vivo™-15 Serum-free Hematopoietic Cell Medium with added M-CSF and IL3. Scale bar = 50µm. (D) Image of microglial cells in maturation culture for two weeks with DMEM/F12 plus nonessential amino acids, glutamine, IL34, CSF1, TGFb2, and CX3CL1. Scale bar = 50µm. (E) Immunohistochemical staining for Iba11 and human CD34, CX3CR1, P2RY12, CD11b, CD68. Scale bar = 100µm. (F) Cell counts and colocalization analysis of (F) CD34- and Iba11- positive cells and (G) positivity for myeloid cell markers CX3CR1, CD11b, activation marker CD68, and microglia marker P2ry12 in differentiated microglia.

expression of *IL6*, *IL1α*, *IL1β*, *TNFα*, *IL8*, *CXCL10*, and *CCL2* mRNA after 6 hours of LPS stimulation (Fig.3B). Similarly, we observed a significant increase in protein expression levels of these cytokines in cell lysate and culture medium (Fig.3C, D). We also treated hiPSC-derived microglia with IFNγ and a combination of IFNγ+LPS (Figure 3. Suppl.), and the results demonstrated that the combination of IFNγ+LPS promoted *IL1α*, *IL1β*, *TNFα*, *CCL8*, and *CXCL10* expression. These findings indicate that hiPSC-derived microglia, akin to microglia in vivo, exhibited strong responses to LPS and IFNγ stimulation.

Microglia are local immune cells in the CNS, functioning as phagocytes involved in clearing apoptotic or necrotic cells, and cell debris (Green et al., 2016), remodeling neuronal connectivity by engulfing synapses, axonal and myelin debris (Paolicelli et al., 2011), and removing pathogens by direct phagocytosis (Nau et al., 2014). To assess the phagocytic capability of hiPSC-derived microglia, we exposed them to three different types of bioparticles: *E. coli* bacteria, zymosan, and bovine photoreceptor outer segments (POS) (Fig.4A-D). The cells altered their morphology in response and rapidly internalized the fluorescent-labeled particles (Fig. 4A-D, Fig.4. Suppl A and B). The engulfed bioparticles were condensed into perinuclear aggregates, likely within lysosomal bodies. They also demonstrated concurrent morphological changes into amoeboid-shaped cells (Fig.4D, E), resembling phenotypes demonstrated by retinal microglia phagocytosing photoreceptors in the context of photoreceptor degenerative pathologies in vivo (Zhao et al., 2015).

Transition of hiPSC-derived microglial cells to a homeostatic state within the mouse retina following xenotransplantation in vivo

To assess the ability of hiPSC-derived microglia to serve as microglia donor cell sources for transplantation, we conducted xenotransplantation experiments using humanized immunodeficient Rag2^{-/-};IL2rg^{-/-};hCSF1^{+/+} transgenic mice as recipients as previously employed (Svoboda et al., 2019; Xu et al., 2020; Chadarevian et al., 2023). Prior to xenotransplantation, recipient transgenic mice were pharmacologically depleted of endogenous retinal microglia by systemic administration of the CSF1R inhibitor PLX-5622 to create a depleted tissue niche for the integration of xenotransplanted microglia (Zhang et al., 2018). Two days following PLX-5622 treatment, adult transgenic mice were transplanted with 5000 hiPSC-derived microglia via injection into the subretinal space (Fig.5A). Tissue analysis at 4 and 8 months post-transplantation revealed that transplanted cells, which were marked by either tdTomato or EGFP expression, had migrated anteriorly from the subretinal space into the neural retina and were distributed across a wide retinal area within the inner and outer retinal layers including the ganglion cell layer (GCL), inner plexiform layer (IPL), and outer plexiform layer (OPL) (Fig.5B-D) in retinal loci typically occupied by endogenous microglia. Transplanted cells were immunopositive for IBA1, and human CD11b (hCD11b) (Fig.5H, I), as well as for microglia signature markers hP2RY12 and hTMEM119 (Fig.5D). Interestingly, the transplanted cells within the retina showed a ramified morphology typical of homeostatic microglia and demonstrated a regularly tiled “mosaic” distribution in their soma positions. These integrated cells were juxtaposed alongside residual endogenous murine microglia, which showed similar morphology and distribution as in endogenous conditions. This indicated that transplanted hiPSC-derived microglia responded to similar intraretinal cues and inter-microglia neighbor-neighbor interactions that guide the spatial and morphological organization of retinal microglia in vivo (Fig.5E-G). Similar observations were made for separate experiments involving the transplantation of tdTomato- and EGFP-expressing hiPSC-derived microglia (Fig.5B,C,H,I).

We further evaluated the impact of human iPSC-derived microglia xenotransplantation on the host mouse retinal cells (Fig. 5 Suppl A-D). We examined the effect of microglia transplantation on endogenous Müller cell morphology and gliosis markers as retinal microglia have been described to interact with Müller glia to regulate overall retinal neuroinflammatory response

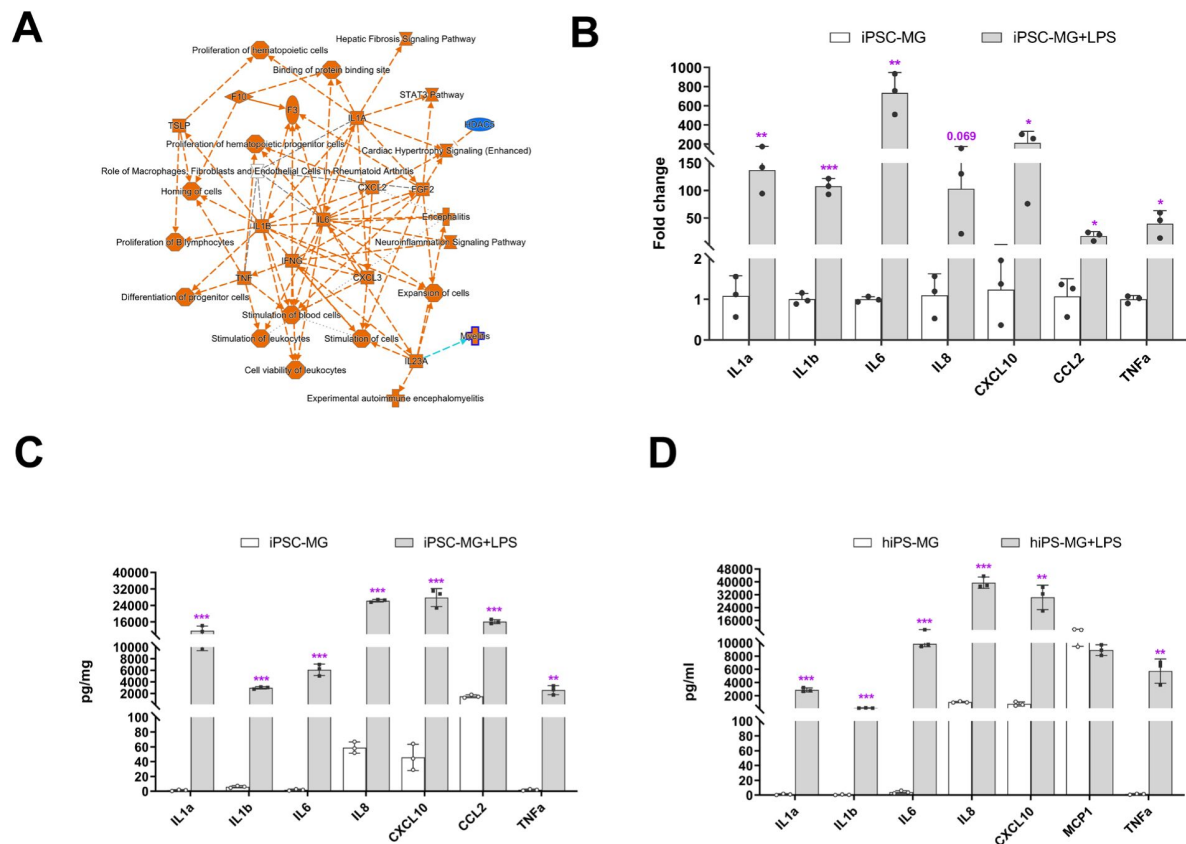


Figure 3.

Inflammation responses of hiPSC-derived microglial cells following Lipopolysaccharide (LPS) stimulation.

(A) Ingenuity Pathway Analysis (IPA) differentially-expressed genes (fold change >2-fold, p < 0.05) between LPS-treated and control hiPSC-derived microglial cells., demonstrating activation of core pathways involving IL6, IL1a, IL1b, and IFNG. (B) Assessment of mRNA expression of selected genes for inflammatory cytokines using QRT-PCR demonstrated increased expression following LPS (0.1µg/ml) stimulation for 6hrs. These changes corresponded to increases in the protein expression levels of inflammatory cytokines following 24 hours of LPS stimulation as measured with a Multiplex kit (Millipore) on celllysate (C) and in conditioned media (D). The data in (C) and (D) are presented as means ± SEM. * p<0.05, ** P<0.01, *** P<0.001, **** P<0.0001.

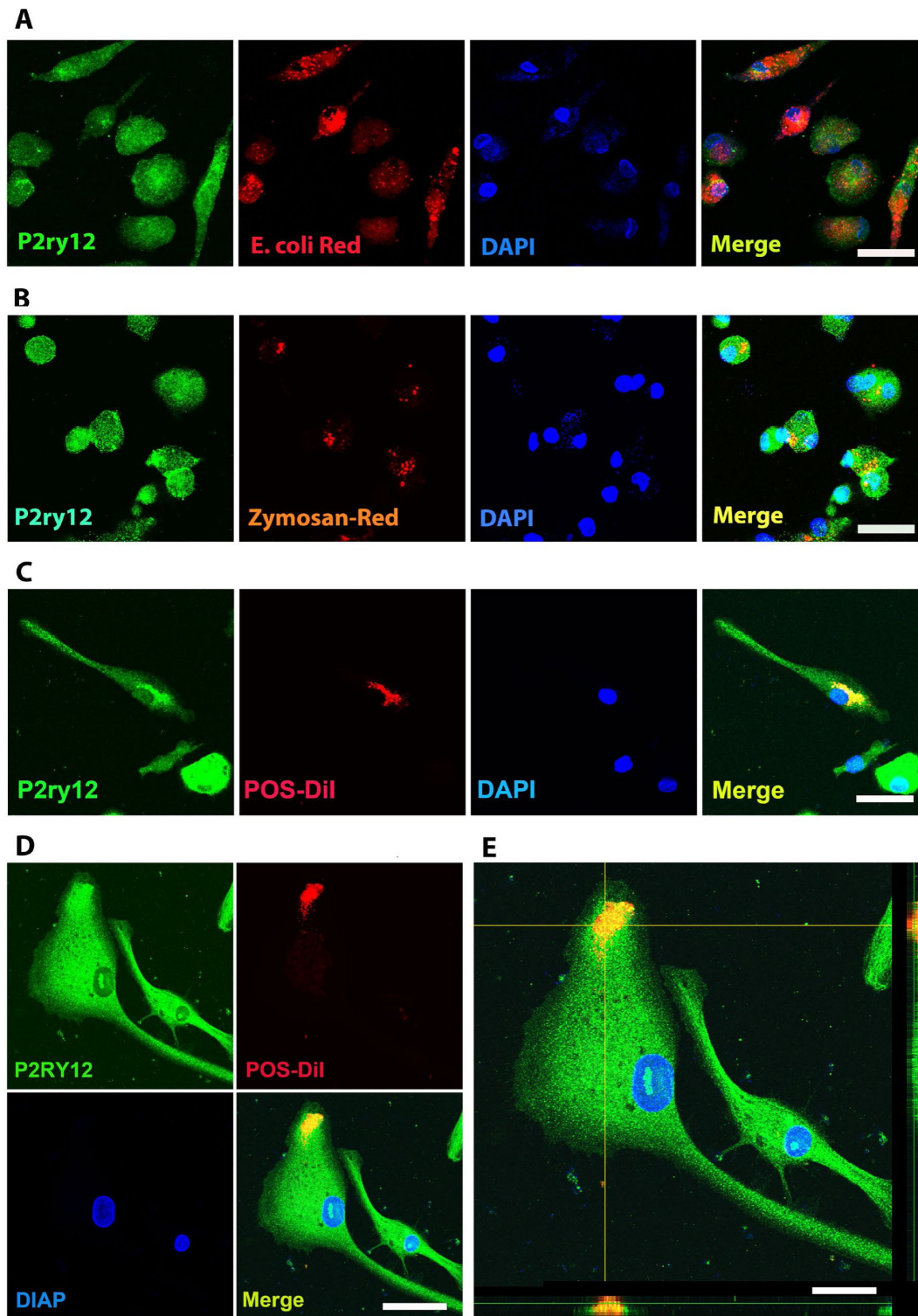


Figure 4.

Human iPSC-derived microglia demonstrate robust phagocytosis.

(A) pHrodo™ Red E. coli bioparticles were incubated in Human iPSC-derived microglia were incubated for 1 hour in either labelled pHrodo™ Red E. coli (A), pHrodo™ Red zymosan bioparticles (B), DiI-labeled bovine photoreceptor outer segments (POS) (C), and colabeled with anti-human P2RY12 antibody (green) and DAPI. Scale bar = 40µm. (D) A high-magnification view of a POS-containing intracellular vesicle within a labeled microglial cell is shown. Scale bar = 30µm.

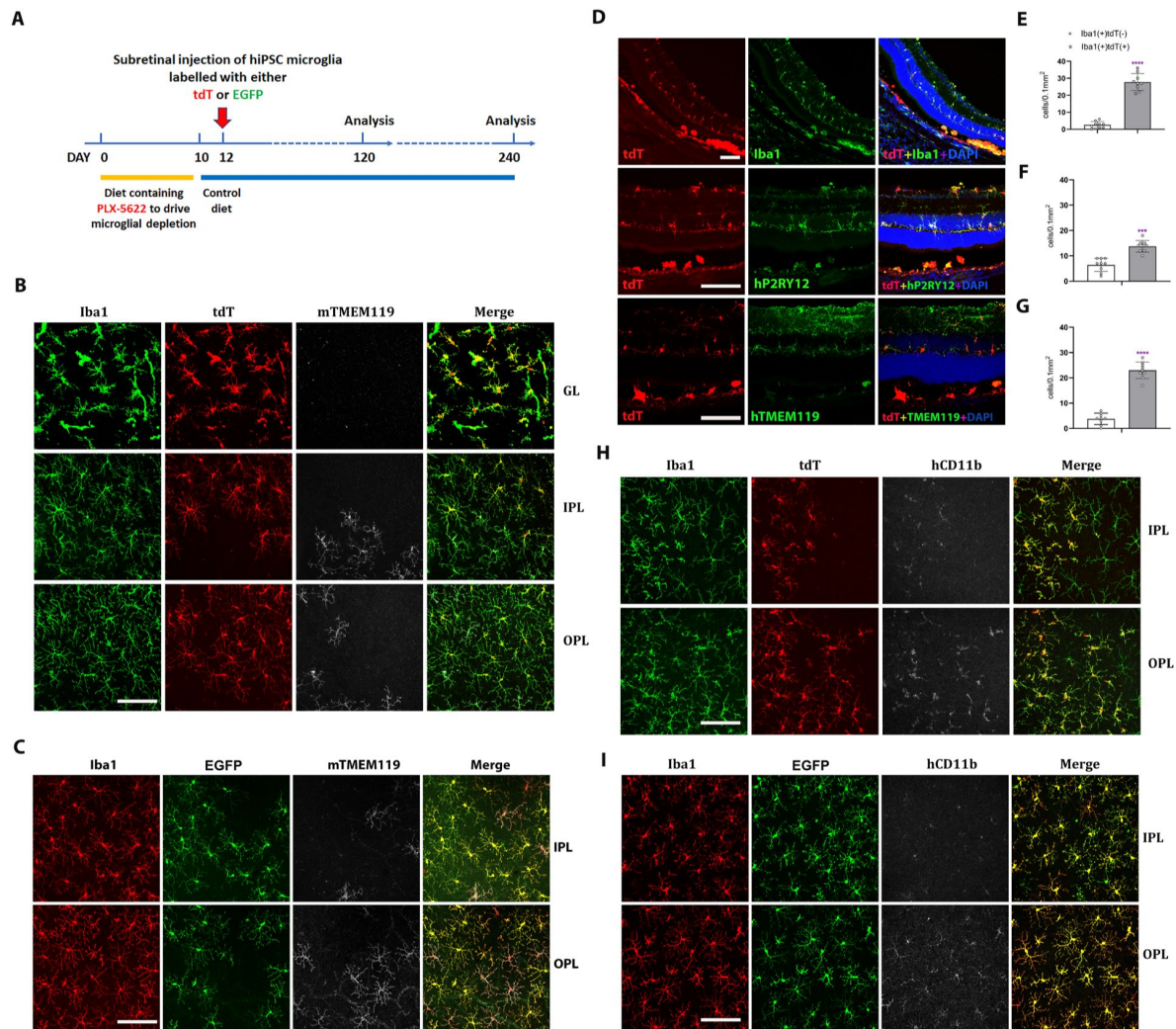


Figure 5.

Human iPSC-derived microglial cells xenotransplanted into recipient mouse retina in vivo demonstrate recapitulation of endogenous distribution, cellular morphology, and stable integration for up to 4 months.

(A) Schematic diagram shows the timeline for transplantation experiments. Two-month old adult transgenic Rag2^{-/-};IL2rg^{-/-};hCSF1^{+/+} mice were fed a PLX-5622-containing diet for 10 days before switching to standard chow. Two days following the resumption of standard chow, human iPSC-derived microglial cells expressing either tdTomato or EGFP were xenotransplanted into the subretinal space via subretinal injection (5000 cells in 1μl injection volume). Retinas were harvested for analysis 120 and 240 days following transplantation. (B) and (C) Retinas isolated post-transplantation were analyzed in flat-mounted tissue with confocal imaging. Transplanted hiPSC-derive microglia were visualized through their expression of tdtomato (TdT) (B) or EGFP (C), while endogenous mouse microglia were visualized using immunostaining for mouse Tmem119 (mTmem119). Imaging analysis was performed in separate layers of the retina including the ganglion cell layer (GL), inner plexiform layer (IPL), and outer plexiform layer (OPL). Scale bar = 100μm. (D) The retinal section showed human iPSC-derived microglial cells integrated into whole retinal layers (top panel) and positively stained with human P2ry12 and TMEM119 microglia signature markers. Scale bar = 100μm. The microglia cell number in GL, IPL, and OPL of host mouse retina were counted: mouse microglial cells (Iba11+, tdT-) and grafted human microglial cells (Iba11+,tdT+) were shown in (E), (F) and (G), respectively. *** P<0.001, **** P<0.0001. (H) and (I) showed tdT (H) or EGFP (I) labeled human iPSC-derived microglial cells in the IPL and OPL of the flat-mount retina with human CD11b staining. These results demonstrated that the infiltration of grafted hiPSC-derived microglial cells into the mouse retina is general in nature and not cell-line specific. Scale bar = 100μm.

(Wang et al., 2014 [↗](#)). Immunostaining with glial fibrillary acidic protein (GFAP) and glutamine synthetase (GS) antibody (**Fig. 5 Suppl. A, B** [↗](#)) indicated that the transplanted human iPSC-derived microglial cells did not trigger any upregulation of Müller cell gliosis markers or induced morphological changes or reactive proliferation for up to four months post-transplantation. Additionally, the laminar organization of the inner and outer retinal layers remained normal and similar to those in control retinas not subjected to transplantation (**Fig. 5 Suppl. A-D** [↗](#)), indicating that microglia transplantation had no adverse impact on the structural integrity of retina. Furthermore, immunostaining with the mouse CD11b antibody which marked the residual population of endogenous mouse microglia showed the spatial juxtaposition of these mouse microglia with transplanted human iPSC-derived microglia in a common mosaic of tiled cells (**Fig. 5 Suppl. E, F** [↗](#)), indicating the ability of transplanted hiPSC-derived microglia to integrate with and exchange signals with the residual endogenous microglial population, but not replaced.

Transplanted hiPSC-derived microglia respond to induction of RPE cell injury with migration and proliferation

To monitor the longer-term consequences of microglia xenotransplantation, we extended analysis and examined the location and morphology of transplanted hiPSC-microglia up to 8 months post-transplantation. Analysis of retinal flat mounts confirmed that the tdTomato⁺ cells remained appropriately located in the GCL, IPL, and OPL, forming a mosaic-like distribution typical of endogenous microglia (**Fig. 6 Suppl. A, B** [↗](#)). These cells maintained expression of hP2RY12 and hTMEM119, markers of homeostatic retinal microglia, indicating their long-term integration within the recipient mouse retina (**Fig. 6 Suppl. A, B** [↗](#)).

To evaluate the function of the transplanted hiPSC-derived microglia in the mouse retina and their ability to respond to injury *in vivo*, we subjected recipient mice 240 days post-transplantation to sodium iodate (NaIO₃)-mediated RPE injury and analyzed retinal tissue 3 and 7 days post-injury (**Fig. 6A** [↗](#)). In a previous study (Ma et al., 2015), we had characterized the responses of endogenous retinal microglia in this injury model; in the few days following injury, microglia within the neural retina migrated into the subretinal space, coming into close proximity to damaged RPE cells. This resulted in a transient decrease in microglia number in the IPL and OPL, which then recovered partially following the proliferation of the remaining microglia to replenish the depleted areas. We found that transplanted hiPSC-derived microglia demonstrated responses similar to endogenous microglia. Three days after NaIO₃ injury, there was an increase in tdTomato⁺, hP2ry12⁺, hiPSC-derived microglia in the subretinal space, while their number decreased in the IPL and OPL (**Fig. 6B-D, G** [↗](#)), indicating a migration of these cells from the neuroretina to the subretinal space. Some of the remaining tdTomato⁺ and P2ry12⁺ cells in the inner retina were positive for the cell-proliferation marker Ki67, indicating active cell division (**Fig. 6B-D, F** [↗](#)). The numbers of Ki67⁺ tdTomato⁺ microglia peaked at three days post-injury and decreased thereafter (**Fig. 6B-F** [↗](#)). Seven days post-injury, the numbers of tdTomato⁺ and P2ry12⁺ human iPSC-derived microglia increased in the IPL and OPL but decreased in the subretinal space (**Fig. 6E, G** [↗](#)); Ki67⁺ tdTomato⁺ cell numbers also declined (**Fig. 6E, F** [↗](#)). These findings suggest that once the hiPSC-derived microglia had replenished endogenous numbers in the inner retina, their division ceased. This response mirrors that of the endogenous mouse retinal microglia to NaIO₃ injury (Ma et al., 2015).

Overall, these results demonstrate that the transplanted human iPSC-derived microglial cells retained a capacity for migration and proliferative responses to injury in a manner observed for endogenous microglia.

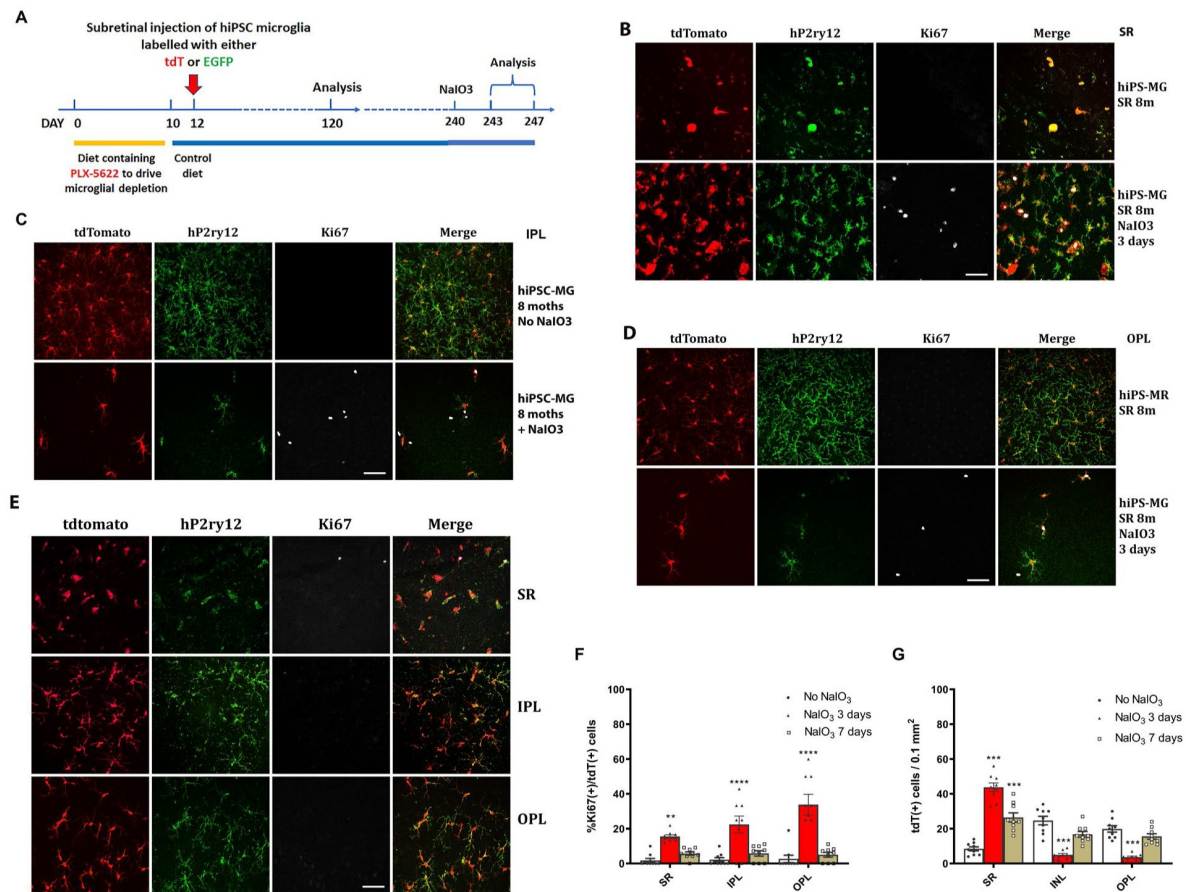


Figure 6.

Migration and proliferation of hiPSC-derived microglia in the mouse retina after sodium iodate-induced RPE cell injury.

(A) The schematic diagram shows the procedure of the experiment. After eight months post-transplantation of hiPSC-derived microglia, recipient animals were administered NaIO₃ (30mg/kg body weight, intraperitoneal injection) to induce RPE injury. Retina were harvested at 3 and 7 days after NaIO₃ administration and microglia numbers in the subretinal space monitored in retina and RPE-choroid flat mounts. (B) RPE-choroid flat-mounts demonstrate an increase of hiPSC-derived microglia (tdTomato+ and P2RY12+) in the subretinal space in response to RPE injury. A subset of subretinal microglia labelled for Ki67 indicating active proliferation. Scale bar = 60µm. (C) and (D) showed the number of P2ry12+&tdtomato+ human microglial cells in IPL (C) and OPL (D) decreased; some of them showed Ki67+ staining, Scale bar = 60µm. The cell count results showed in (F) and (G). (E) The retinal flat mount showed the number of P2ry12+&tdtomato+ human microglial cells in IPL and OPL that were repopulated, and the cells stopped dividing with lost the Ki67 staining at seven days after NaIO₃ injection. The cell numbers were shown in (F) and (G). Scale bar = 60µm.

hiPSC-derived microglial cells phagocytize debris or dead photoreceptor cells after NaIO₃-induced RPE cell injury

Phagocytosis, a critical function of microglia, is essential both during development and in the resolution of pathological processes. Retinal microglia adaptively phagocytose and clear apoptotic photoreceptors in the rd10 mouse model of photoreceptor degeneration (Silverman et al., 2019 [DOI](#)). We found here that three days following NaIO₃-induced RPE cell injury, tdTomato+ transplanted hiPSC-derived microglia migrated not only to the subretinal space but also into the photoreceptor layer (**Fig. 7A, C** [DOI](#)), coincident with the time of photoreceptor degeneration, when photoreceptor morphology becomes disrupted and photoreceptor density decreases (**Fig. 7A** [DOI](#)). Within the photoreceptor layer, hiPSC-derived microglia were observed to phagocytose photoreceptors as evidenced by the accumulation of intracellular autofluorescent material in their soma (**Fig. 7A, B, D** [DOI](#)), and their transformation into larger amoeboid cells (**Fig. 7B** [DOI](#), yellow triangle) containing arrestin-immunopositive photoreceptor-derived debris. mRNA analysis for human-specific transcripts also indicated that transplanted hiPSC microglia upregulated inflammatory cytokine expression, increased phagocytosis, and decreased expression of microglia homeostatic genes and neurotrophic factors (**Figure 7. Suppl** [DOI](#), Suppl. Table 5). Taken together, these findings further demonstrate that the xenografted human iPSC-derived microglial cells generate functional responses to in vivo injury that closely resembled those of endogenous homeostatic retinal microglia.

Discussion

Microglial cells are instrumental in the development and progression of numerous CNS diseases. For instance, in Alzheimer's disease, microglia are enriched in over 50% of associated gene loci implicated in AD risk (Hasselmann et al., 2020). Similarly, in Age-related Macular Degeneration, 57% of 368 genes, located close to 52 AMD gene loci, are expressed in retinal microglial cells (**Fig. discussion suppl B** [DOI](#)), and 52% of them are highly expressed in these cells (**Fig. discussion suppl A** [DOI](#) & **B** [DOI](#); Fritsche et al., 2016 [DOI](#); Ma et al., 2013 [DOI](#); Den et al., 2022 [DOI](#)). Understanding microglia cell functions is essential for investigating disease mechanisms and identifying accurate therapeutic targets. Most of our current knowledge about microglial cells comes from rodent studies. However, genetic and functional differences exist between murine and human microglia (Galatro et al., 2017 [DOI](#); Gosselin et al., 2017 [DOI](#); Smith & Dragunow, 2014 [DOI](#)). Hence, more in-depth knowledge of human microglial cells in vitro and in vivo is required.

Human-induced pluripotent stem cells (iPSC) offer promising prospects for many retinal research fields (Zhong et al., 2014 [DOI](#); Leach et al., 2016 [DOI](#); Tanaka et al., 2016 [DOI](#)). For over a decade, macrophage/microglial cells have been differentiated using human iPSC (Karlsson et al., 2008 [DOI](#); Pocock et al., 2018 [DOI](#)). An abundance of hiPSC-derived microglial cells, with defined genomic background, and easy manipulation of hiPSC offer substantial benefits in various research areas.

Under in vivo physiological conditions, microglial cells exhibit a tile-like arrangement, without overlap. They try to maintain this property in culture, allowing overgrown cells to float out to the medium. Based on this phenomenon and a variety of established microglia cell differentiation protocols (Muffat et al., 2016 [DOI](#); Pandya et al., 2017 [DOI](#); Abud et al., 2017 [DOI](#); Douvaras et al., 2017 [DOI](#); Haenseler et al., 2017 [DOI](#); Takata et al., 2017 [DOI](#)), We chose to employ the myeloid progenitor/microglia cell floating culture method (described in Van et al., 2013, and Haenseler et al., 2017 [DOI](#)) due to its simplicity, efficiency, and consistency. This approach facilitates the generation of a large and uniform population of myeloid progenitor cells, with over 98.6% expressing the CD34+ marker, sustained over a three-month period. These progenitor cells differentiate into pure microglial cells (>98.5% P2ry12+), bearing a profile rich in microglia genes, and demonstrate characteristics similar to native microglia in physiological CNS tissue. A

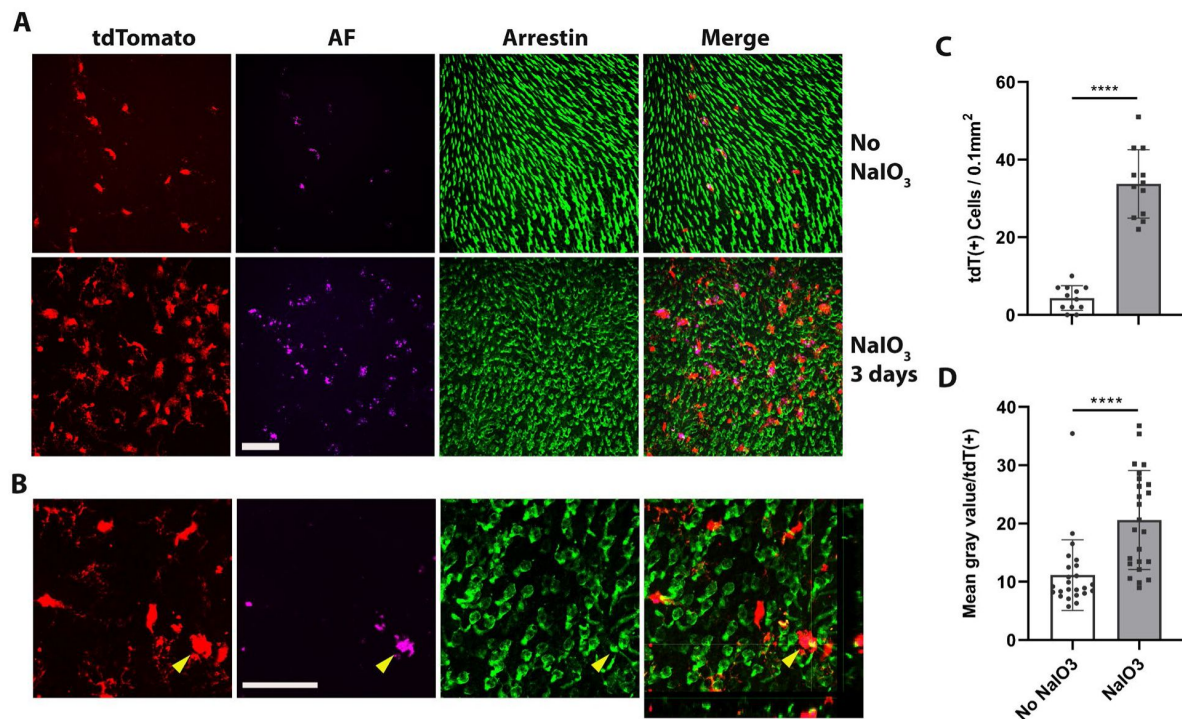


Figure 7.

Dyshomeostasis human iPSC-derived microglial cells in the mouse retina phagocytose dead photoreceptor cells/debris after RPE cell injury.

(A) Dyshomeostasis human microglial cells (tdtomato+) accumulated in the photoreceptor cell layer after 3 days of NaIO₃-induced RPE cell injury compared with no NaIO₃ administration. The photoreceptor cells stained with cone arrestin (green), autofluorescent showed in magenta. Scale bar = 60μm. (B) High magnification and side view image showed human microglial cells (red) co-labeled with photoreceptor cells arrestin staining (green) after 3 days of NaIO₃ injury. The yellow triangle showed the colocalized tdT+ human microglia cell and arrestin+ cone photoreceptor cell. Scale bar = 40μm. (C) The number of human microglial cells in the photoreceptor layer; (D) The mean gray value of autofluorescence in each human microglia cell. **** P<0.0001.

comparison of the signaling pathway to myeloid cells reveals the central hubs of signaling as IL6, IL1b, and the stat3 pathway (**Fig.2D** [↗](#)), which are key to microglia functioning during inflammation. The differentiation protocol employing floating myeloid progenitor cells produces a significant number of CNS resident-like microglial cells. These hiPSC-derived microglial cells respond robustly to LPS stimulation and demonstrate phagocytic activity, mimicking primary cultured retinal microglial cells. Gene expression profile comparison between hiPSC-microglial cells and fetal/adult brain microglia revealed that hiPSC-derived microglial cells are comparable to fetal and adult microglial cells but are far away from monocytes and inflammatory monocytes (**Fig.2. Suppl** [↗](#). B, C and D; Suppl. table 1 and table 2). Therefore, they effectively replicate resident microglia characteristics (Ulland et al., 2018 [↗](#); Wang et al., 2020 [↗](#); Shi et al., 2022 [↗](#); Guo et al., 2022 [↗](#)).

CNS disease treatment strategies include gene regulation, gene delivery (Neumann et al., 2006 [↗](#); Beutner et al., 2013 [↗](#)), rejuvenation (Karlstetter et al., 2015 [↗](#); Elmore et al., 2018 [↗](#)), or replacement of dysfunctional microglial cells (Willis et al., 2020 [↗](#); Xu et al., 2020 [↗](#); Han et al., 2020 [↗](#); Shibuya et al., 2022 [↗](#)). hiPSC-derived microglial cells offer a potentially unlimited source for cell replacement therapy. The in vivo functionality of these cells was verified through xenografting into adult Rag2^{-/-}; IL2rg^{-/-};hCSF1^{+/+} mice. The grafted microglial cells integrated into the right location of the retina, expressing microglia signature genes and formed new homeostasis with resident microglial cells for eight months. Homeostatic hiPSC-derived microglia responded to RPE cell injury like host retinal microglial cells, marking the success of the xenotransplantation model (Guilliams et al., 2017 [↗](#)) and highlighting the potential for hiPSC-derived microglial cells in retinal microglia cell replacement therapy, such as in AMD.

Several exogenous microglia cell replacement techniques have been investigated, with these methods varying based on the type of donor cells used and the age of the recipient. Initial microglia cell replacement studies started from the transplantation of hematopoietic stem cells (HPSC) (Larochelle et al., 2016 [↗](#); Xu et al., 2020 [↗](#); Hohsfield et al., 2020 [↗](#)). However, even after differentiation in local tissue, HPSC-derived microglial cells maintained some gene expressions distinct from the original resident microglial cells (Lund et al., 2018 [↗](#)), warranting further exploration of these HPSC-derived microglial cells' unique characteristics. Another method involves the use of newborn mice as recipients, with iPSC or stem cells-derived microglial cells as the donors (Mancuso et al., 2019 [↗](#); Xu et al., 2020 [↗](#)). This approach does not require the depletion of resident microglial cells, and the grafted cells can infiltrate the brain tissue, distributing similarly to the original resident microglial cells. While suitable for examining microglia cell function in various backgrounds, the clinical applicability of this method remains limited.

A third transplantation approach involves adult recipients receiving iPSC or stem cell-derived microglial cells after resident microglia cell depletion (Chadarevian et al., 2023 [↗](#)). Since this technique requires a microglial-empty niche, resident microglia must be depleted or relocated from the retina, typically done by using CSF1R inhibitors. To prevent these inhibitors from affecting the grafted cells, investigators have tried using modified CSF1R microglial cells as donor cells (Chadarevian et al., 2023 [↗](#)). In our research, we found a two-day recovery period with normal food intake sufficient to clear the CSF1R inhibitor, allowing grafted cells to integrate into microglia empty niche. In this study, we used the subretinal xenotransplantation method, though alternative transplantation routes such as intravenous injection, vitreous, and suborbital space delivery warrant further exploration.

In conclusion, hiPSCs can be differentiated into microglial cells through a simplified common pathway and factors, although the more precise differentiation factors still need further investigation under in vitro conditions. For instance, the microglia signature gene, TMEM119, shows low expression in hiPSC-derived microglia single-type cell culture conditions. Interestingly, we discovered that a medium composed of mix cultured embryoid bodies and microglia

progenitor cells fosters further differentiation of microglial cells. The humanized mouse model established through xenotransplantation serves as a reliable tool for studying the functions of various hiPSC-derived microglial cells. This model offers a valuable platform for investigating disease mechanisms and evaluating the therapeutic effects of xenografted hiPSC-derived microglial cells from variety of backgrounds.

Experimental procedures

Experimental animals and PLX-5622 treatment

In vivo experiments were conducted according to protocols approved by the Institutional Animal Care and Use Committee (National Eye Institute Animal Care and Use Committee) and adhered to the Association for Research in Vision and Ophthalmology (ARVO) statement on animal use in ophthalmic and vision research. Rag2^{-/-};IL2rg^{-/-};hCSF1^{+/+} transgenic mice were obtained from Jackson Laboratories (Stock #17708). Animals were housed in a National Institutes of Health (NIH) animal facility under a 12-h light/dark cycle and fed standard chow. Genotype analysis by sequencing revealed that the rd8 mutation was absent in the *Crb1* gene. To deplete retinal microglia, two-month old experimental animals were administered a diet containing PLX-5622 (Plexxikon, at 1200 parts per million), a potent and selective inhibitor of the CSF1R previously demonstrated to deplete the majority of microglia in the mouse brain (Dagher et al., 2015 [DOI](#)) and retina (Zhang et al., 2018 [DOI](#)). Animals were maintained continuously on the PLX-5622 diet for 10 days and then switched back to standard chow.

Human iPSC Culture

Five separate human iPSC lines were used in this study. The KYOUDXR0109B (201B7, ATCC:ACS1023) hiPSC line was generated in Yamanaka Lab from fibroblasts isolated from a healthy female donor and reprogrammed by the expression of OCT4, SOX2, KLF4 and MYC using retroviral transduction. The lines of NCRM5-AAVS1-CAG-EGFP (clone 5), ND2-AAVS1-iCAG-tdTomato (clone 1), and NCRM6 and MS19-ES-H were obtained from the NHLBI iPSC Core Facility of National Heart, Lung and Blood Institute (NHLBI). NCRM5-AAVS1-CAG-EGFP is an EGFP-expressing reporter iPSC line with CAG-EGFP targeted mono-allelically at the AAVS1 safe harbor locus in NCRM5 iPSCs. These iPSCs were reprogrammed from CD34⁺ PBMCs from a healthy male donor. ND2-AAVS1-iCAG-tdTomato is a tdTomato-expressing reporter iPSC line with insulated CAG-tdTomato targeted mono-allelically at the AAVS1 safe harbor locus in ND2 iPSCs. These iPSCs were reprogrammed using healthy male fibroblast cells. Microglia differentiated from these two reporter lines were used for the xenotransplantation experiments. The NCRM6 iPSC line was reprogrammed from CD34⁺ PBMCs from a healthy female donor with episomal iPSC reprogramming vectors (Thermo Fisher). The MS19-ES-H line was reprogrammed from PBMCs from a healthy female donor with a Cytotune Sendai Virus kit (Thermo Fisher).

The derivation and characterization of iPSC lines were conducted at the ATCC and NHLBI iPSC Core Facility. Cells were cultured on Geltrex-coated (0.2mg/ml, Gibco, #A1413302) 6-well plates using 1x mTeSRTM-1 medium. Passaging was performed following dissociation with TrypLE Express enzymatic digestion (Gibco by Life Technologies). Upon initial plating, cells were cultured in medium containing 3μM Rho-kinase inhibitor (Y-27632, Abcam). The medium was completely refreshed every day. Cells reaching 70% confluence were either passaged or cryopreserved in Stem Cell Freezing Media (mFreSR™, StemCells, Catalog # 05854).

Myeloid progenitor cell differentiation and microglial cell maturation

The protocol for the differentiation of hiPSC cells to myeloid progenitors and then to microglia were adapted and modified from those described previously by the Cowley laboratory (Van et al., 2013, Haenseler et al., 2017 [\[link\]](#)). The first step of the protocol to enable embryoid body (EB) formation employed the Spin-EBs formation method performed in AggreWells™800 microwell culture plates (Stemcell Technologies, Catalog # 34825). Briefly, 1 mL of mTeSR™1 spin-EB medium was added into each culture well and centrifuged at 3000G for 2 minutes. Subsequently, 4×10^6 iPSCs in 1ml of medium were added to the well of spin-EB plate and then centrifuged at 800 rpm for 3 minutes. The plate was incubated at 37°C, 5% CO₂ for four days, then 1 mL medium was replaced in a drop-wise manner every day for the next 4 days with an EB medium: mTeSR1 medium (STEMCELL Technologies) containing 50 ng/mL BMP-4 (GIBCO-PHC9534), 20 ng/mL human stem cell factor (SCF, Miltenyi Biotec), 50 ng/mL vascular endothelial growth factor (hVEGF, GIBCO-PHC9394).

In the second step of myeloid progenitor differentiation, the resulting EBs were harvested with a 40µm filter column. Approximately 150-200 EBs were transferred into a 75-cm² flask containing myeloid cell differentiation medium: TheraPEAK™ X-VIVO™-15 Serum-free Hematopoietic Cell Medium (Lonza, Cat#: BEBP04-744Q) containing 100 ng/mL M-CSF (Invitrogen), 25 ng/mL IL-3 (R&D), 2 mM Glutamax supplement (Invitrogen), 1xN2 supplement (ThermoFisher Scientific, Cat#17502048)). Two-thirds of the media volume in the culture flask was replaced every five days for 2-3 weeks.

In the third step of microglia differentiation, the non-adherent floating cell layer consisting of differentiated myeloid progenitor cells was harvested from the supernatant and transferred into the 6-well plate and allowed to settle and adhere overnight. Two-thirds of the medium volume was removed and replaced with a microglia cell differentiation medium: DMEM/F12 medium (Gibco, #11330) containing 50 ng/mL M-CSF (Invitrogen), 100 ng/mL IL-34 (Peprotech), 10 ng/mL TGFβ1 and 2-5ng/ml TGFβ2 (both from R&D Systems), 20ng/ml CX3CL1 (Peprotech), 1x N2 supplement and 2 mM Glutamax supplement). The culture was maintained for 2 weeks after which the differentiated iPSC-derived microglia were harvested for analysis or for cell transplantation.

Phagocytosis Assay

To assess microglia phagocytosis in vitro, the following bioparticles were employed as targets: 1) pHrodo™ Red E. coli BioParticles Conjugate for Phagocytosis (ThermoFisher Scientific, Cat#P35361); 2) pHrodo™ Red Zymosan Bioparticles™ Conjugate for Phagocytosis (ThermoFisher Scientific, Cat#P35364); 3) bovine rod photoreceptor outer segments (POS, Invision Bioresources, Cat#98740). Bovine POS were diluted in serum-free DMEM/F12 (1:1; Gibco) to a concentration of 10⁶ segments/ml and fluorescently labeled with the lipophilic dye DiI (Vybrant Cell-Labeling Solutions; Invitrogen) according to the manufacturer's instructions. Each phagocytosis assay employed 1x10⁵ POSs and 2mg/ml pHrodo™ Red E. coli membrane/Zymosan BioParticles.

For the assay, harvested floating myeloid cells were transferred into 4-well chamber slides (Thermo Fisher). For phagocytosis assessment of myeloid cells, the cells were cultured for one day and challenged with bioparticles. To assess iPSC-derived microglia, the cells were cultured in a microglia cell differentiation medium for two weeks and then challenged. In the assay, bioparticles were added to the 100µl serum-free DMEM/F12 medium in the slide chamber, incubated for one hour at 37°C, washed 3 times with PBS, and then fixed in 4% paraformaldehyde (PFA) for 20 minutes. Fixed cells were immunostained with antibodies to IBA1 and P2RY12 and counterstained with DAPI. Stained cells were imaged with an Olympus 1000 confocal microscope, and image analysis was conducted using Image J software (NIH).

mRNA and protein analysis following LPS challenge in vitro

Differentiated hiPSC-derived microglia cultured in 6-well plates were stimulated with LPS at 100 ng/mL for 6 and 24 hr. Microglia stimulated for 6 hrs were collected in RNeasy lysis solution (Thermo Fisher) and stored at -80°C for further QRT-PCR analysis. Microglia stimulated for 24 hrs were collected for protein quantification. After the medium was collected, the cells in the well were washed with 1XPBS, and then 200µl of RIPA lysis buffer with proteinase inhibitor cocktail (Calbiochem) was added; the cells were removed by scraping, collected into 1.5ml Eppendorf tube, and then homogenized with sonication (Sonicator 125 Watts, Qsonica) at 4°C. After sonication and centrifugation, total protein concentration was measured (BCA protein assay kit; Pierce). Levels of individual cytokines were determined using a Milliplex bead assay kit (Milliplex MAP human cytokine/chemokine magnetic bead panel, #MCYTOMAG-70K; Millipore) and involving the Luminex MAPIX system with data analysis using xPONENT 4.2 software (Luminex). The cytokines analyzed included *IL1α*, *IL1β*, *IL6*, *IL8*, *TNFα*, *CXCL10*, *CCL2*, *CCL3*, *CCL4*, *IL10*.

mRNA expression analysis by quantitative RT-PCR

mRNA expression was quantitated using quantitative reverse transcription-PCR (qRT-PCR). Harvested cells were lysed by trituration and homogenized using QIAshredder spin columns (Qiagen). Total RNA was isolated using the RNeasy Mini kit (Qiagen) according to the manufacturer's specifications. First-strand cDNA synthesis from mRNA was performed using qScript cDNA SuperMix (Quanta Biosciences) using oligo-dT as primer. qRT-PCR was performed using an SYBR green RT-PCR kit (Affymetrix), using the Bio-Rad CFX96 Touch™ Real-Time PCR Detection System under the following conditions: denaturation at 95 °C for 5 min, followed by 40 cycles of 95 °C for 10 s, and then 60 °C for 45 s. Threshold cycle (CT) values were calculated and expressed as fold-induction determined using the comparative CT ($2^{\Delta\Delta CT}$) method. Ribosomal protein S13 (RPS13) and GAPDH were used as internal controls. Oligonucleotide primers are provided in Suppl. Table 4.

Transplantation of hiPSC-derived microglia by subretinal injection

Differentiated hiPSC-derived microglia grown in flasks were first washed in 1XPBS before being removed by scraping and collected into a 50ml tube in 5ml PBS. Cell numbers were counted using a cell counter (Countess 3, ThermoFisher). Microglia were collected by centrifugation (5 minutes at 4°C, 200G) and the resulting cell pellet resuspended in 1XPBS at a concentration of 5000 cells/µl for in vivo transplantation via subretinal injection. Experimental animals were given general anesthesia (ketamine 90 mg/kg and xylazine 8 mg/kg) and additional topical anesthesia (0.5% Proparacaine HCL, Sandoz) applied to the injected eye. For the injection, the temporal sclera was exposed by a conjunctival cut-down and a scleral incision made 0.5mm behind the limbus using 32G needle to access the subretinal space. The tip of a blunt 32G needle attached to a Hamilton micro-syringe was introduced through the incision at an angle 5-degrees tangent to the globe and advanced 0.5-1 mm into the subretinal space under a dissecting microscope. Microglial cells (5000 cells in 1µl PBS) were slowly injected from the micro-syringe into the subretinal space using an aseptic technique. Post-procedure, treated eyes were carefully inspected for signs of bleeding or distention and intraocular pressure was monitored using a tonometer (iCare TONOLAB, Finland). In the event that intraocular pressure remained elevated (>20mmHg) and/or the globe appeared distended, a vitreous tap was performed using a 33-G needle to reduce intraocular pressure. In the unlikely event that excessive bleeding is observed, the animal will be examined by a veterinarian or euthanized immediately.

Immunohistochemical analyses

For immunohistochemical analysis of microglia in vitro, microglia were differentiated in 4-well chambered slides, fixed in 4% PFA for 20 minutes and processed for immunostaining. For in vivo analyses, recipient animals were euthanized by CO₂ inhalation, and the eyes were enucleated.

Enucleated eyes were dissected to form posterior segment eyecups and fixed in 4% paraformaldehyde in phosphate buffer (PB) for 2h at 4°C. Eyecups were either cryosectioned (Leica CM3050S) or further dissected to form retinal flat mounts. Flat-mounted retinas were blocked for 1h in a blocking buffer containing 10% normal donkey serum and 1% Triton X-100 in PBS at room temperature. Primary antibodies included IBA1 (1:500, Wako, #019–19741), anti-mouse Tmem119 (1:500, Synaptic Systems, #400 004), anti-human TMEM119 (1:100, Sigma, #HPA051870), anti-mouse Cd68 (1:200, BioRad, #MCA1957), anti-human CD68 (1:100, R&D, #MAB20401), anti-mouse Cd45 (1:100, Bio-Rad, #MCA1388), anti-human CD45 (1:100, R&D, #FAB1430R), cone arrestin (1:200, Millipore, #AB15282), Ki67 (1:30, eBioscience, #50-5698-82), anti-P2RY12 (1:100, ThermoFisher, #PA5-77671 and Sigma, #HPA014518), CD34 (1:50, eBioscience, #14-0341), hCD11b (1:100, R&D, #FAB1699R), mCD11b (BioRad, Cat#: MCA711G, 1:100), CX3CR1 (1:100, Invitrogen, #61-6099-42), hHLA (Invitrogen, #11-9983-42), PU.1 (Invitrogen, #MA5-15064), TREM2 (1:100, Invitrogen, #702886), glutamine synthetase (1:200, Millipore, #MAB302), PKCa (1:200, Sigma-Aldrich, #p4334), GFAP (1:200, Invitrogen, #13-0300), RBPMs (1:100, Phosphosolutions, 1832-RBPMs), Calbindin (1:5000, Swant, CB-38a), and anti-RFP (1:200, RockLand, 600-401-379-RTU). Primary antibodies were diluted in blocking buffer and incubated at 4°C overnight for retinal sections and at room temperature overnight on a shaker for retinal flat mounts. Experiments in which primary antibodies were omitted served as negative controls. After washing in 1xPBST (0.2% Tween-20 in PBS), retinal samples were incubated for 2 hrs at room temperature with secondary antibodies (AlexaFluor 488-, 568- or 647-conjugated anti-rabbit, mouse, rat, goat and guinea pig IgG) and DAPI (1:500; Sigma-Aldrich) to label cell nuclei. Isolectin B4 (IB4), conjugated to AlexaFluor 568/647 (1:100, Life Technologies), was used to label activated microglia and retinal vessels. Stained retinal samples were imaged with confocal microscopy (Olympus FluoView 1000, or Zeiss LSM 880, or Nikon A1R). For analysis at high magnification, multiplane z-series were collected using 20 or 40 objectives; Confocal image stacks were viewed and analyzed with FV100 Viewer Software, NIS-Element Analysis and Image J (NIH).

RNAseq analysis

For whole transcriptome analysis, cultured myeloid progenitor cells (MPC) and differentiated microglial cells with and without 0.1µg/ml LPS treatment were harvested in the flasks and 6-well plates, respectively. After harvesting, all samples stored frozen in RNeasy (Roche) solution before RNA extraction using the Qiagen RNA Mini Kit. RNA quality and quantity were evaluated using Bioanalyzer 2100 with the RNA 6000 Nano Kit (Agilent Technologies). The preparation of RNA library and transcriptome sequencing was performed using an external vendor (Novogene, Sacramento, CA). Genes with adjusted p-value <0.05 and log2FC (Fold Change) > 1 were considered differentially expressed. Ingenuity Pathway Analysis (IPA, Qiagen) was employed for canonical pathway and graphical pathway analysis. The microglia gene list was constructed from our previous microarray data from retinal microglial cells and published data (Ma et al. 2013 [\[1\]](#); Bennett et al., 2016). The heat map, volcano, and histogram plot were performed using Prism 9.5.1 (Graph-Pad).

Sodium iodate (NaIO₃)-induced model of RPE cell injury

Recipient mice 8 months following xenotransplantation were administered a single dose of NaIO₃ (Honeywell Research Chemicals) at a dose of 30 mg/kg body weight via intraperitoneal injection. Animals were euthanized 3 and 7 days after NaIO₃ injection, and their retinas harvested and subjected to histological and molecular analysis.

Statistics and reproducibility

All data in the graphical panel represent mean ± standard error. When only two independent groups were compared, significance was determined by a two-tailed unpaired t-test with Welch's correction. When three or more groups were compared, one-way ANOVA with the Bonferroni post

hoc test or two-way ANOVA was used. A P value <0.05 was considered significant. The analyses were done in GraphPad Prism v.5. All experiments were independently performed at least three experimental replicates to confirm consistency in observations across replicates.

Supplemental information

Supplemental Information includes supplemental figures and legends and supplemental tables.

Author contributions

WM performed and analyzed data and wrote the manuscript; WT and WL designed and supervised the studies and edited the manuscript. LZ and BX performed in vivo procedures, RNF assisted with imaging procedures and collection, TMR supervised and supported the study, and JZ provided the hiPSC lines and iPSC quality control.

Acknowledgements

This study was supported by the National Eye Institute Intramural Research program.

Fig 1. Suppl.

Immunocytochemistry staining with human PU.1 and TREM2(A). Scale bar = 100µm. (B) PU.1 and TREM2 positive cells are 99.7% and 99.4%, respectively, in entire DAPI+ cell counts.

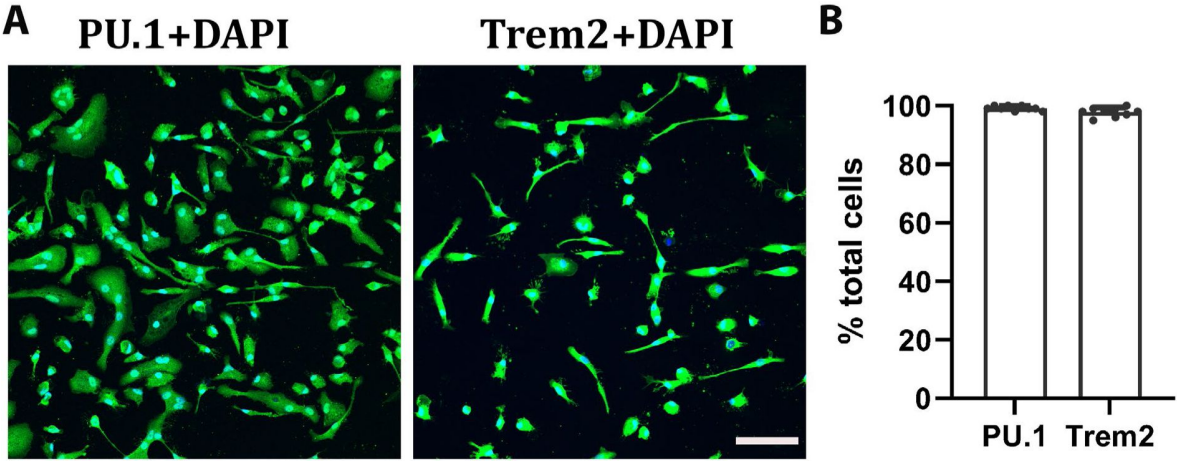
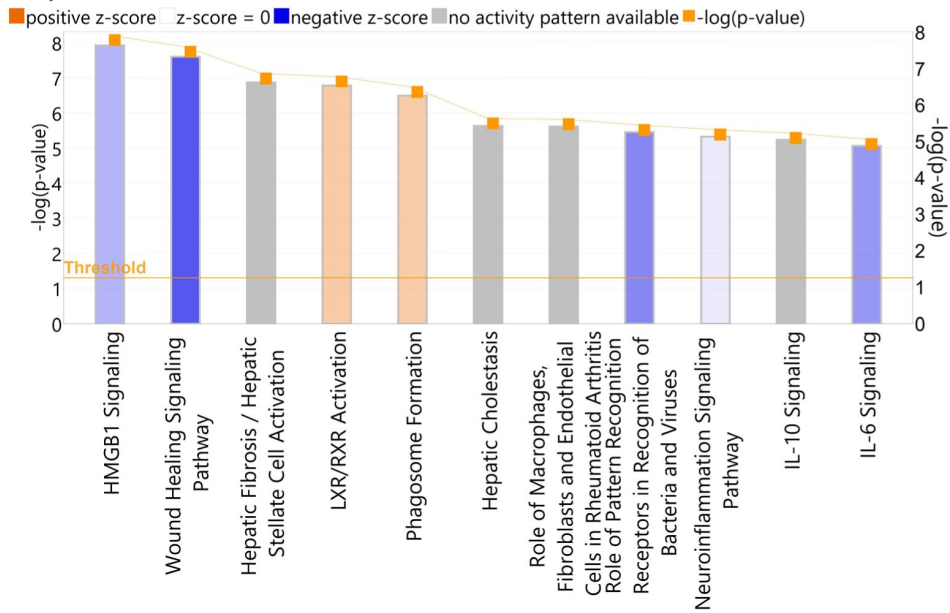


Fig 2. Suppl A.

Analysis results of the canonical pathway of completed differentiated hipsc-derived microglia vs. myeloid progenitor cells.



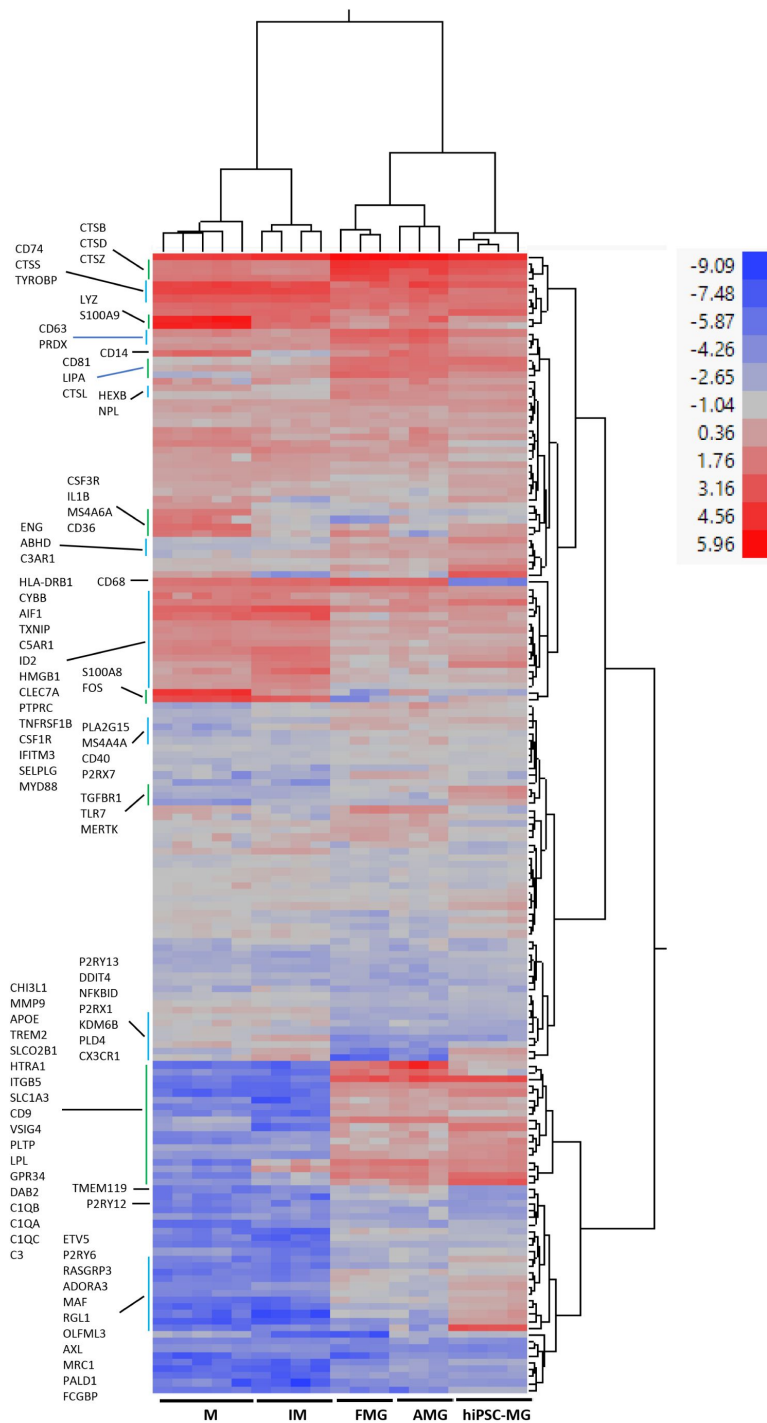


Fig 2. Suppl B.

Hierarchical cluster analysis on microglia-enriched genes among hiPSC-derived microglial cells (hiPSC-MG) and human adult brain microglia cells (AMG), fetal brain microglia cells (FMG), inflammatory monocytes (IM), monocytes (M). The human microglia gene panel combined our mouse microglia-enriched genes and human microglia-enriched genes (Abud et al., 2017; Muffat et al., 2016; Douvaras et al., 2017; Böttcher et al., 2019; Van et al., 2019). The total microglia-enriched gene list contains 203 genes (Suppl. Table 1), which were used to be extracted from the gene profile of hiPSC-MG and downloaded human adult microglia (AMG), fetal microglia (FMG), inflammatory monocyte (IM) and monocytes (M) (GSE 178846, Abud et al., 2017). 188 genes (Suppl. table 1) were obtained from both gene profiles. All the gene counts were normalized with 4 human cell housekeeping genes C1orf43, RAB7A, REEP5 and VCP (Eisenberg et al., 2013). The hierarchical cluster was analyzed by JMP (JMP Statistical Discovery LLC.). Results showed that hiPSC-MG is more comparable with AMG and FMG.

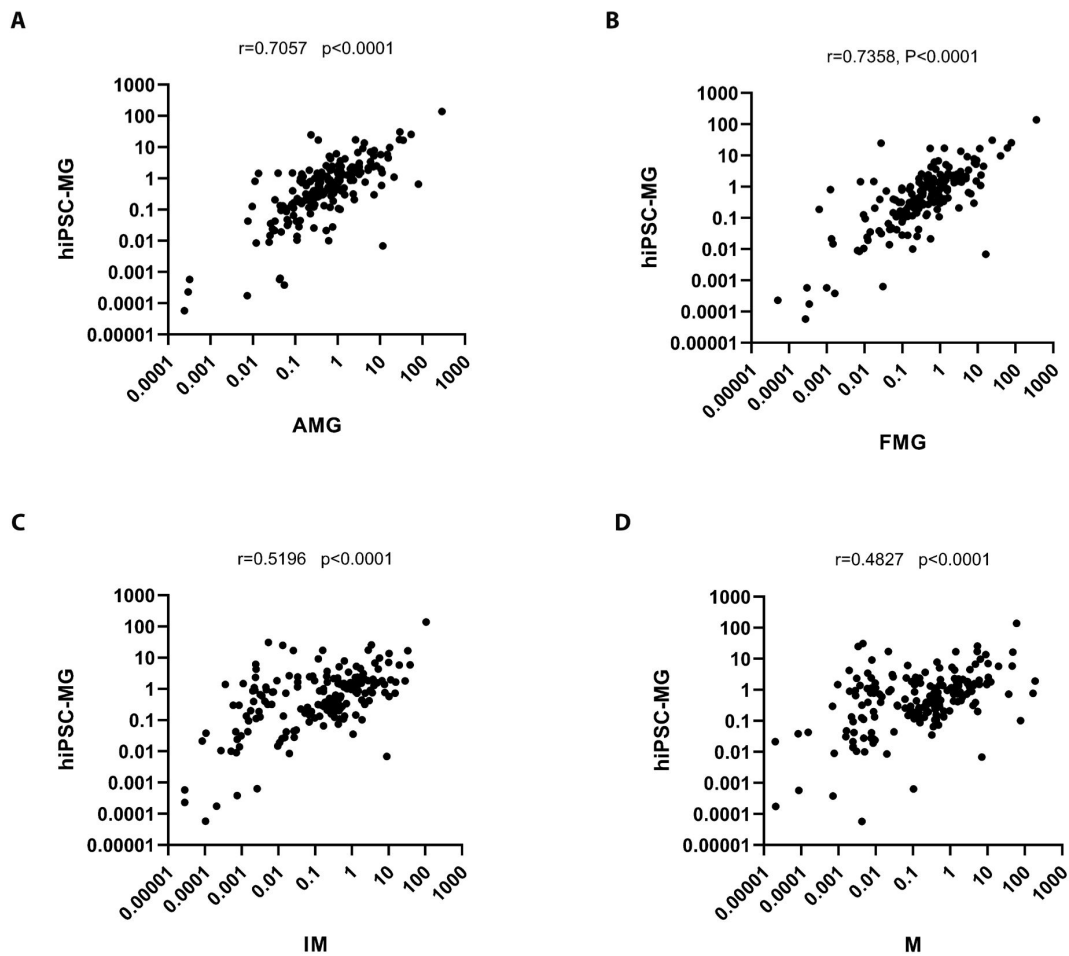


Fig 2. Suppl C.

The correlation analysis of human microglia enriched genes among hiPSC-derived microglia cells (hiPSC-MG), human adult brain microglia cells (A), fetal brain microglia cells (B), inflammatory monocytes (C), monocytes (D) respectively. The images and the analysis results (Prism, GraphPad) showed hiPSC-MG correlated significantly with FMG ($r=0.7358$, $p<0.0001$) and AMG ($r=0.7057$, $p<0.0001$).

Fig 2. Suppl D.

Correlation comparison of entire gene profiles between male (A)/female (B) hiPSC-derived microglial cells and male (A)/female (B) human brain microglia cells (GSE 111972, [van der Poel et al. 2019](#) [DOI](#); Suppl. table 2) respectively. The results showed they are significantly correlative ($r=0.8055(M)$, $r=0.8326(F)$, $P<0.0001$).

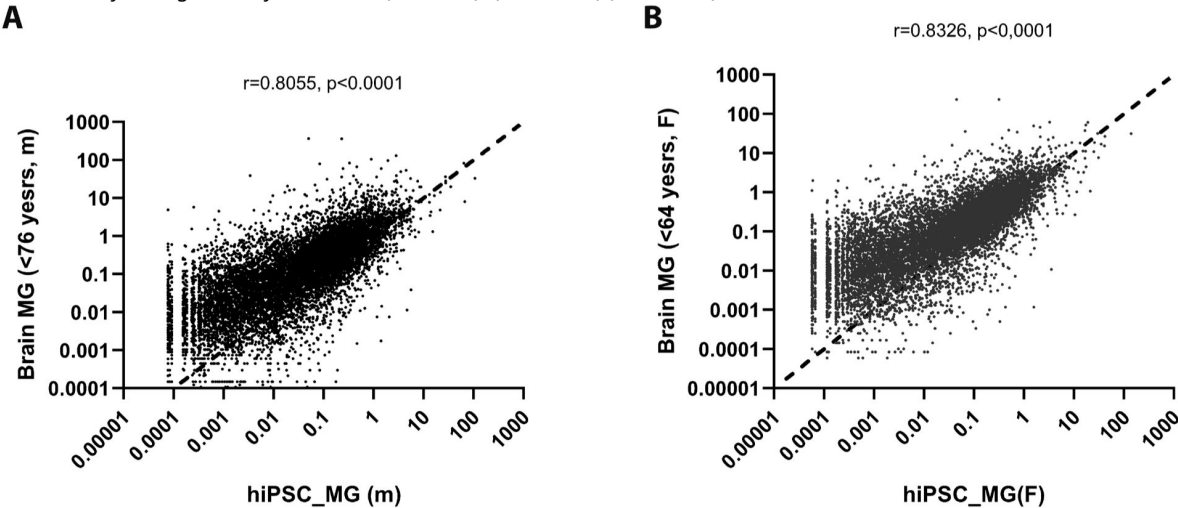
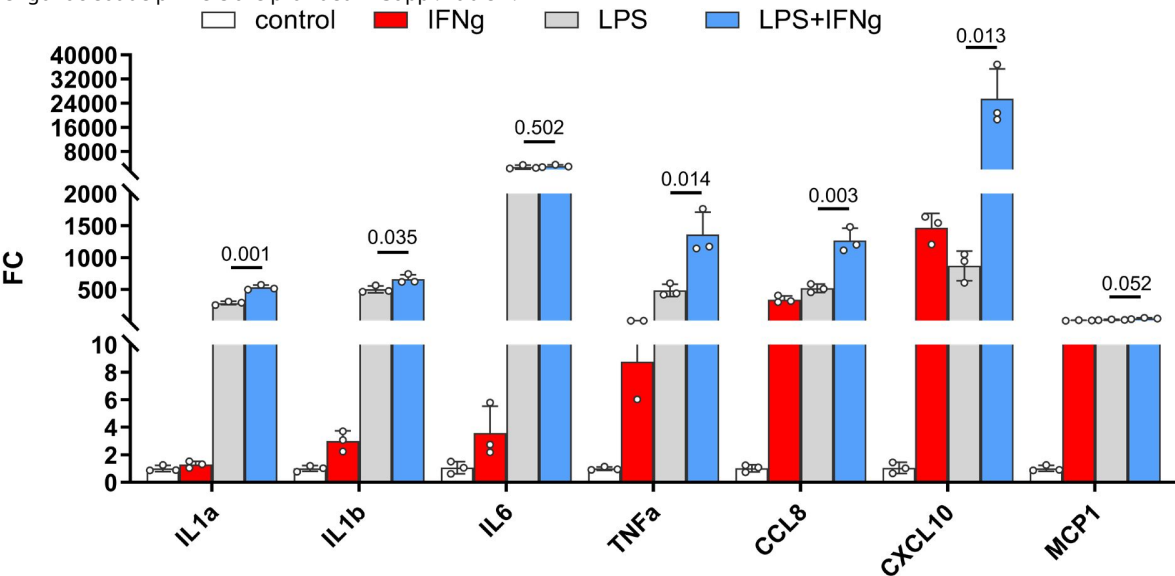


Figure 3 suppl.

Inflammatory cytokines were produced synthetically by IFN γ and LPS in hiPSC-derived microglial cells. hiPSC-derived microglia cells were cultured in 6-well plates for 14 days, 20ng/ml of human IFN γ and 0.5ug/ml of LPS were added to the wells, respectively, or 20ng/ml of human IFN γ and 0.5ug/ml of LPS were added to the wells together. After 6 hours of incubation, the cells were washed and harvested into a 1.5ml Eppendorf tube for RNA isolation with a NucleoSpin RNA/Protein Mini kit (Macherey-Nagel, #740933.50). The cDNA was synthesized with PrimeScript™ 1st strand cDNA Synthesis Kit (Takara, #6110A). qRT-PCR was performed using a SYBR green RT-PCR kit (Affymetrix), using the Bio-Rad CFX96 Touch™ Real-Time PCR Detection System under the following conditions: denaturation at 95 °C for 5 min, followed by 40 cycles of 95 °C for 10 s, and then 60 °C for 45 s. Threshold cycle (CT) values were calculated and expressed as fold-induction determined using the comparative CT ($2^{-\Delta\Delta CT}$) method. Ribosomal protein S13 (RPS13) and GAPDH were used as internal controls. Oligonucleotide primers are provided in Suppl. Table 4.



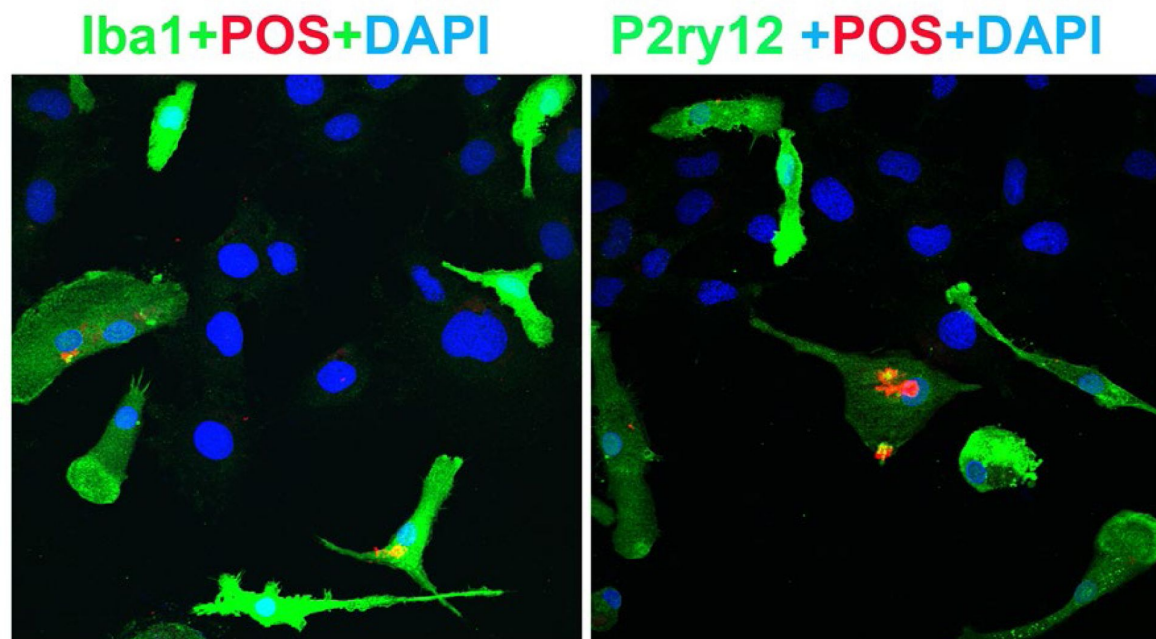


Fig 4. Suppl A.

Floating myeloid progenitor cells were cultured in a 2-well slide chamber overnight, and then the DiI-labeled bovine photoreceptor outer segments were added to the chamber and incubated for 1 hr. The cells were fixed with 4% PFA for 20 minutes and stained with Iba1 and hP2ry12, respectively. The DiI-labeled bovine photoreceptor outer segments can only be engulfed by Iba1 or hP2ry12-positive microglia cells. Scale bar = 16 μ m.

Zymosan treatment for 1hr

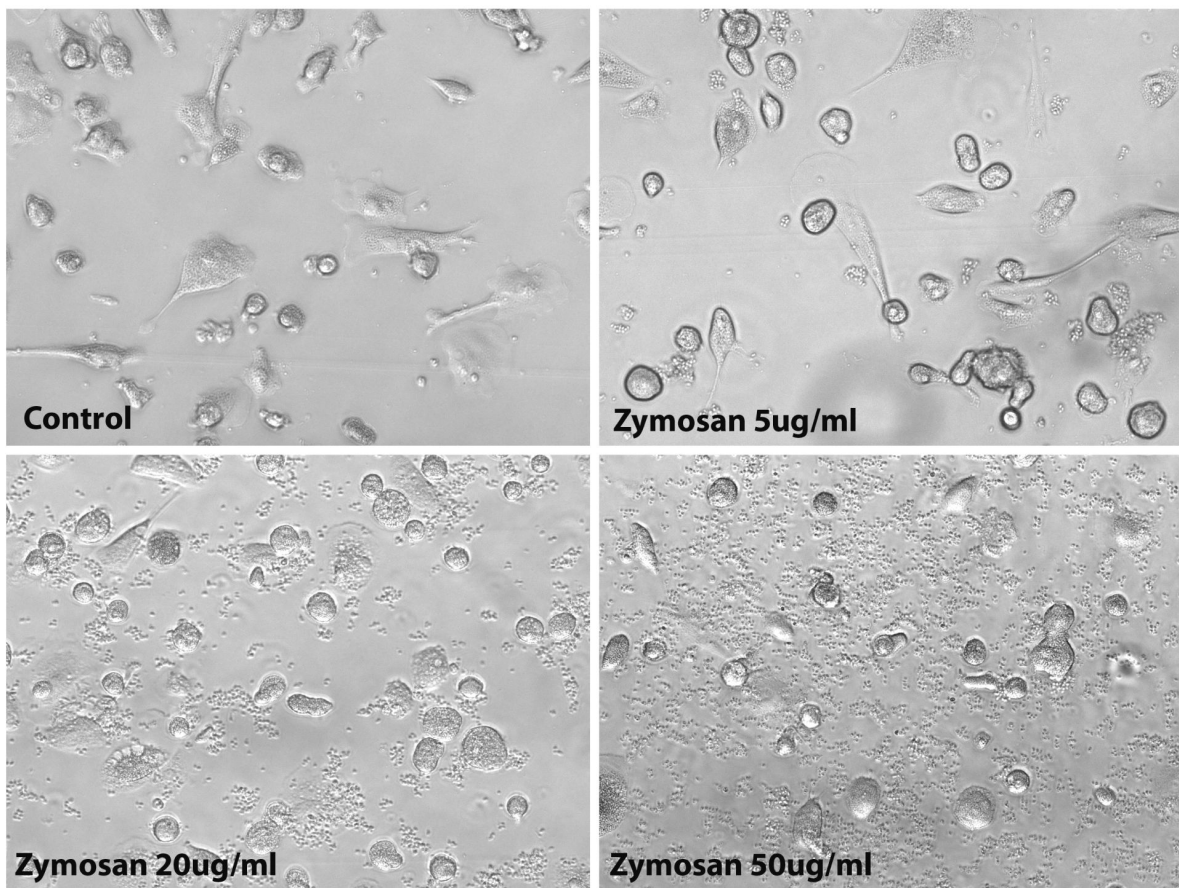


Figure 4. Suppl B.

The morphology of hiPSC-derived microglia cells under 1-hour zymosan treatment with different concentrations. After 1 hour of treatment with zymosan, 5ug/ml concentration didn't change the microglia cell morphology, but over 20ug/ml concentration changed the microglia morphology to an ameboid round shape.

Fig. 5. Suppl A.

Homeostatic hiPSC-derived microglial cells in the mouse retina do not affect local retinal cells. Entire section images showed GFAP, GS, RBPMS, and Iba1 staining for astrocytes, Müller cells, ganglion cells, and microglia cells in the retina after four months of xenotransplantation. Scale bar = 300µm.

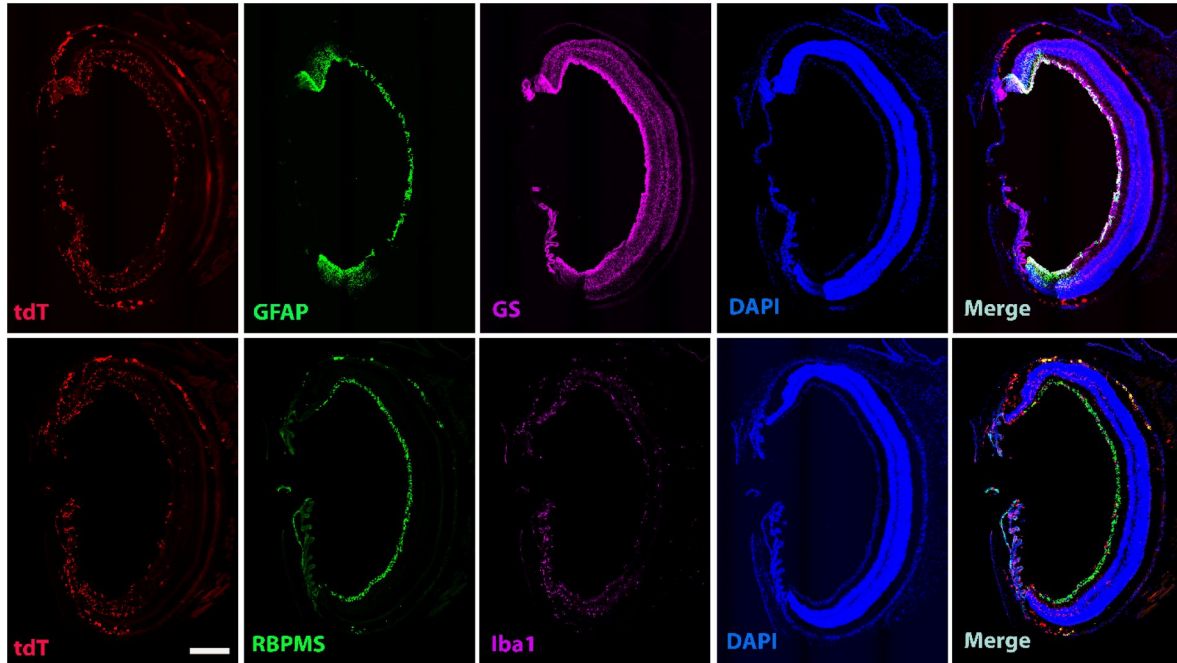
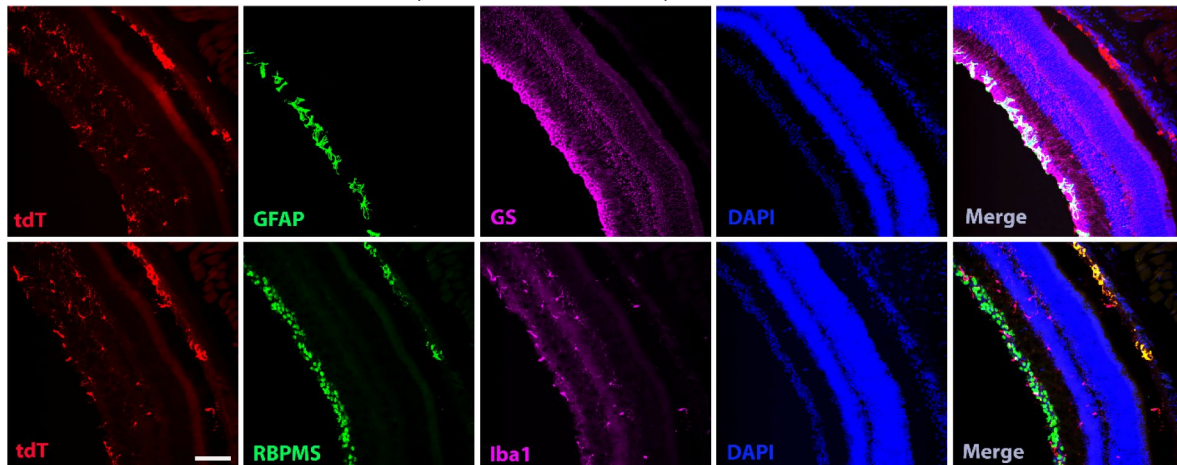


Fig. 5. Suppl B.

Homeostatic human iPSC-derived microglia cells in the mouse retina do not affect local retinal cells. The partial section high magnification images showed GFAP, GS, RBPMS, and Iba1 staining for astrocytes, Müller cells, ganglion cells, and microglia cells in the retina after four months of transplantation. Scale bar = 100µm.



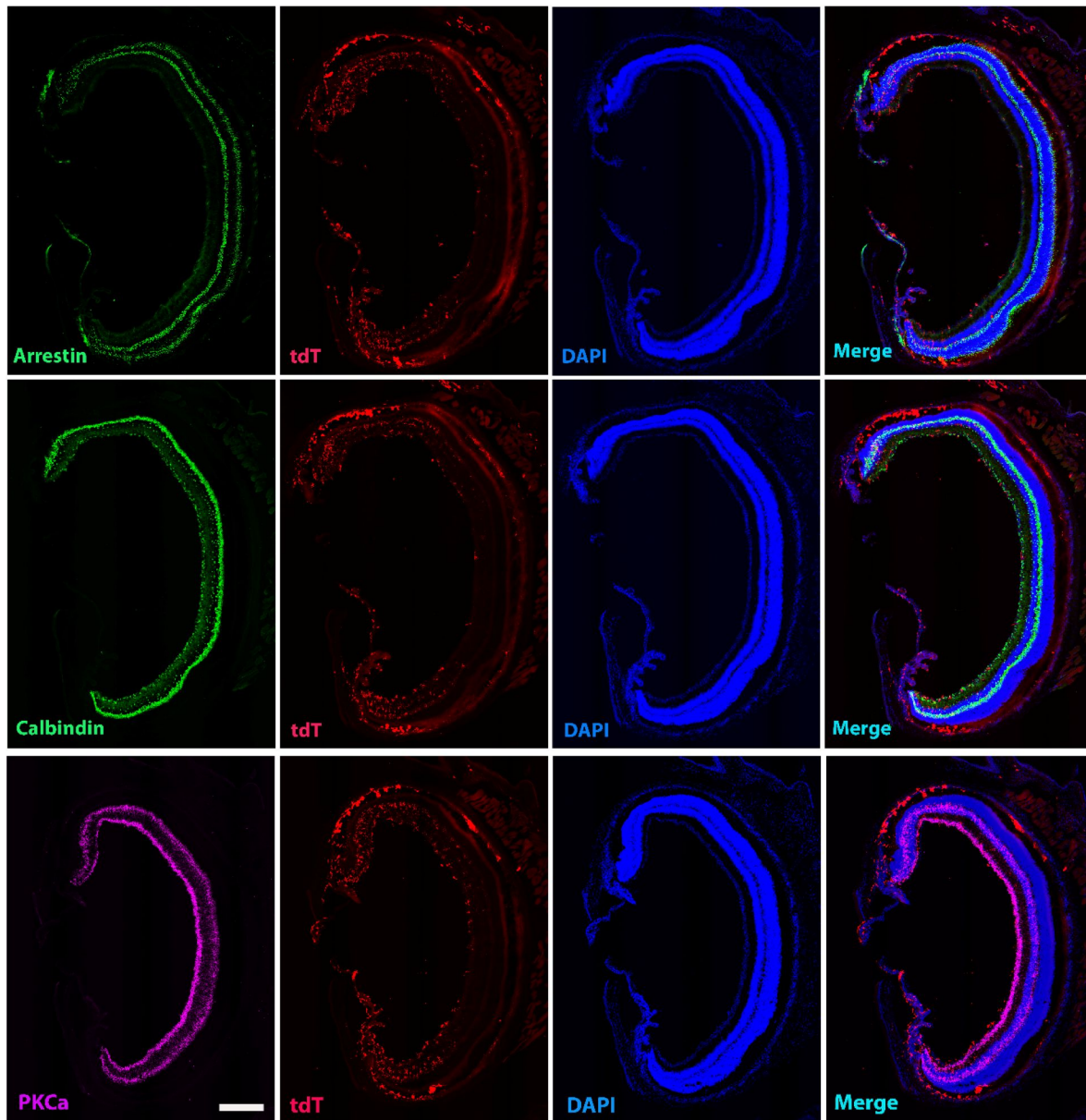


Fig. 5. Suppl C.

Homeostatic human iPSC-derived microglia cells in the mouse retina do not affect local retinal cells. Entire section images showed arrestin, calbindin, and PKCa staining for cone photoreceptors, horizontal and some amacrine cells, and bipolar cells in the retina after four months of xenotransplantation. Scale bar = 300µm.

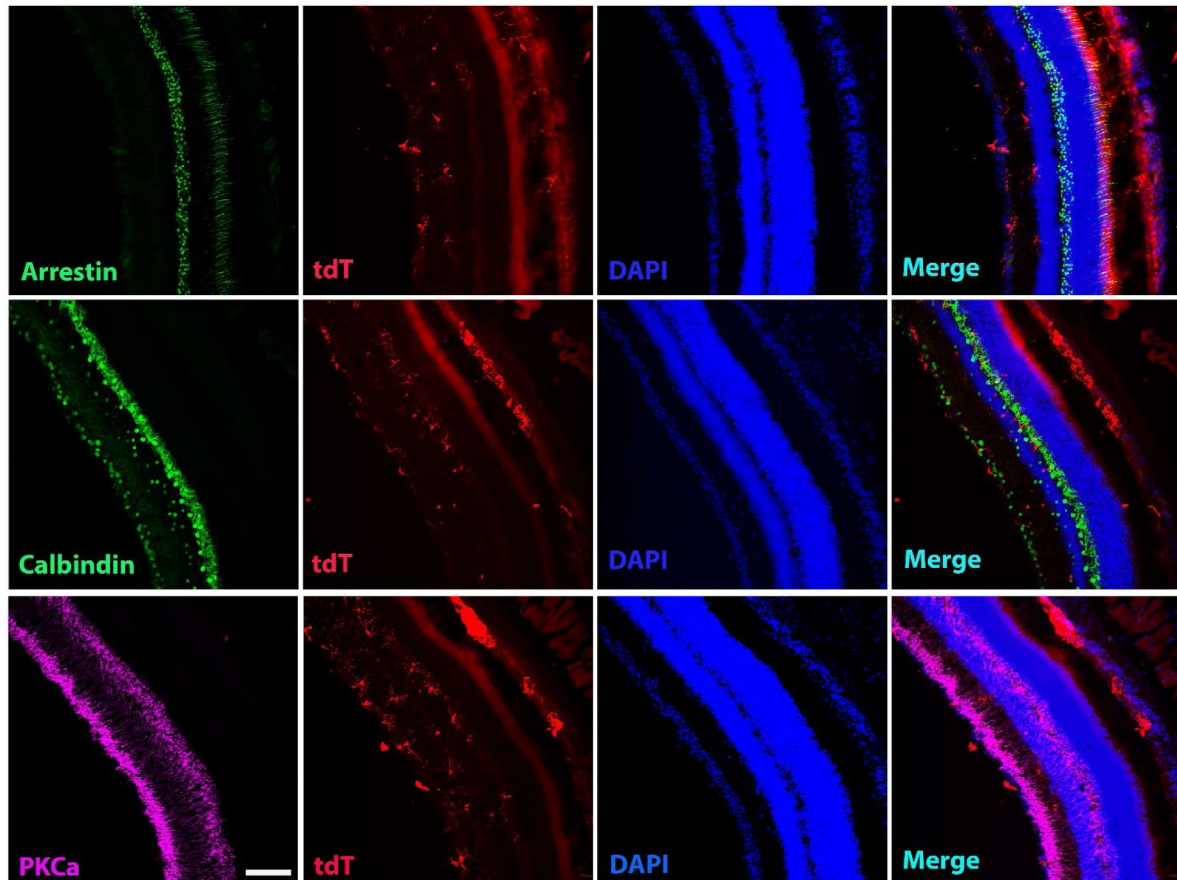


Fig. 5. Suppl D.

Homeostatic human iPSC-derived microglia cells in the mouse retina do not affect local retinal cells. The partial section of high magnification images showed arrestin, calbindin, and PKCa staining for cone photoreceptors, horizontal and some amacrine cells, and bipolar cells in the retina after four months of xenotransplantation. Scale bar = 100 μ m.

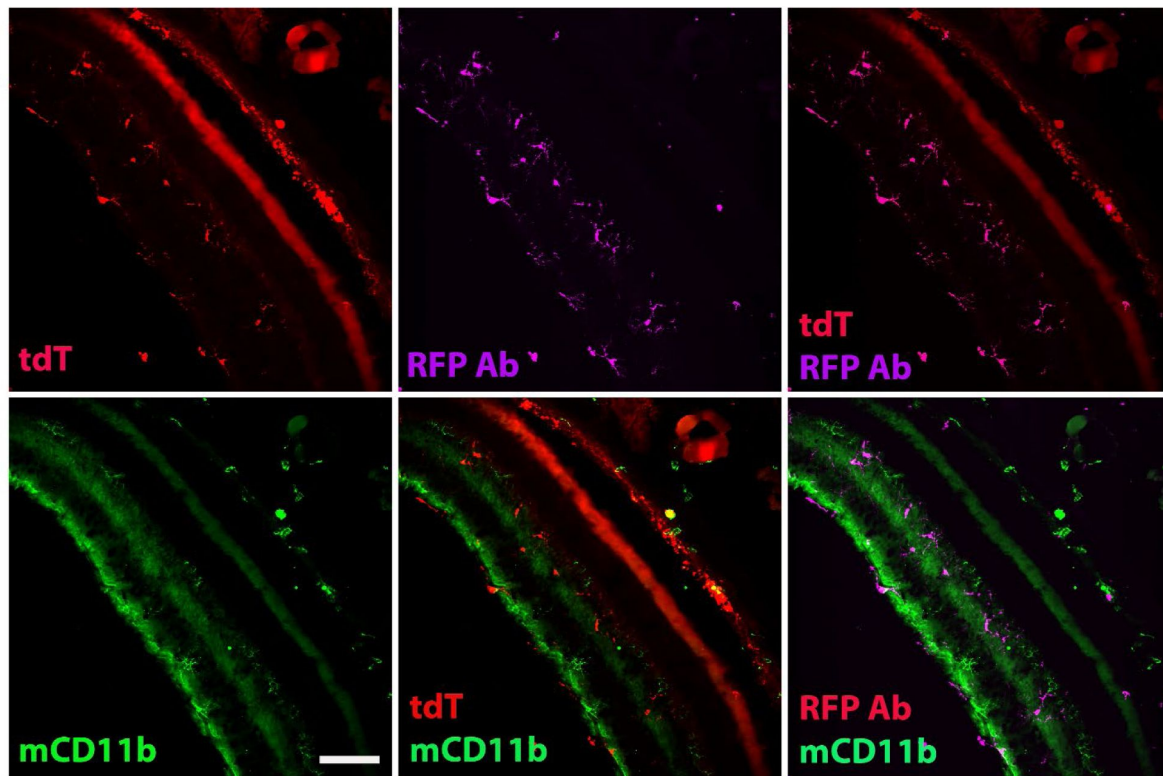


Fig. 5. Suppl E.

Homeostatic human iPSC-derived microglia cells in the mouse retina do not take over local retinal microglia cells. The section images showed RFP and mouse CD11b staining to determine the tdT+ human microglia cells and local mouse microglia cells in the retina after four months of xenotransplantation. The results showed that the tdT+ human cells colocalized with RFP staining (Far red) but not with mouse CD11b (green) staining. Scale bar = 100µm.

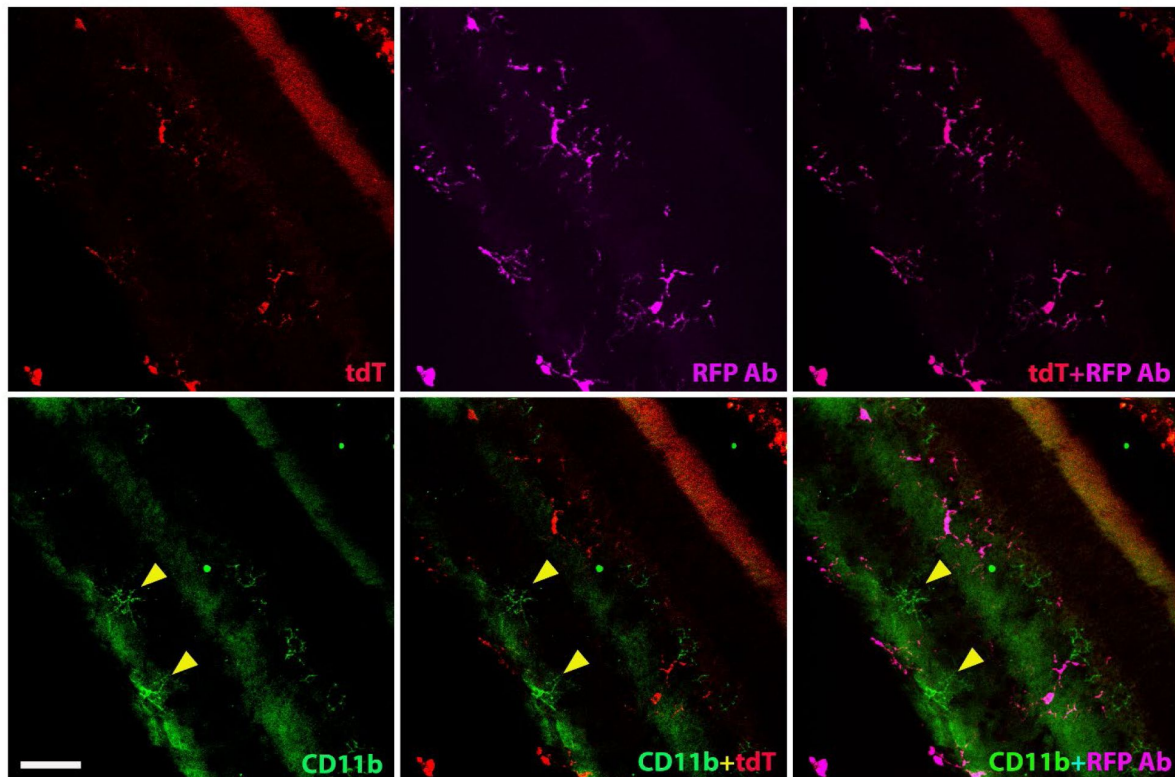


Fig. 5. Suppl F.

Homeostatic human iPSC-derived microglia cells in the mouse retina do not take over local retinal microglia cells. The magnification images showed RFP and mouse CD11b staining in the retina after four months of xenotransplantation. The results showed that the tdT+ human cells colocalized with RFP staining (Far red) but not with mouse CD11b(green) staining. The triangle marker indicated that the local mouse microglia cells were only stained with mouse CD11b but not colocalized with tdT+ human microglia cells. Scale bar = 50µm.

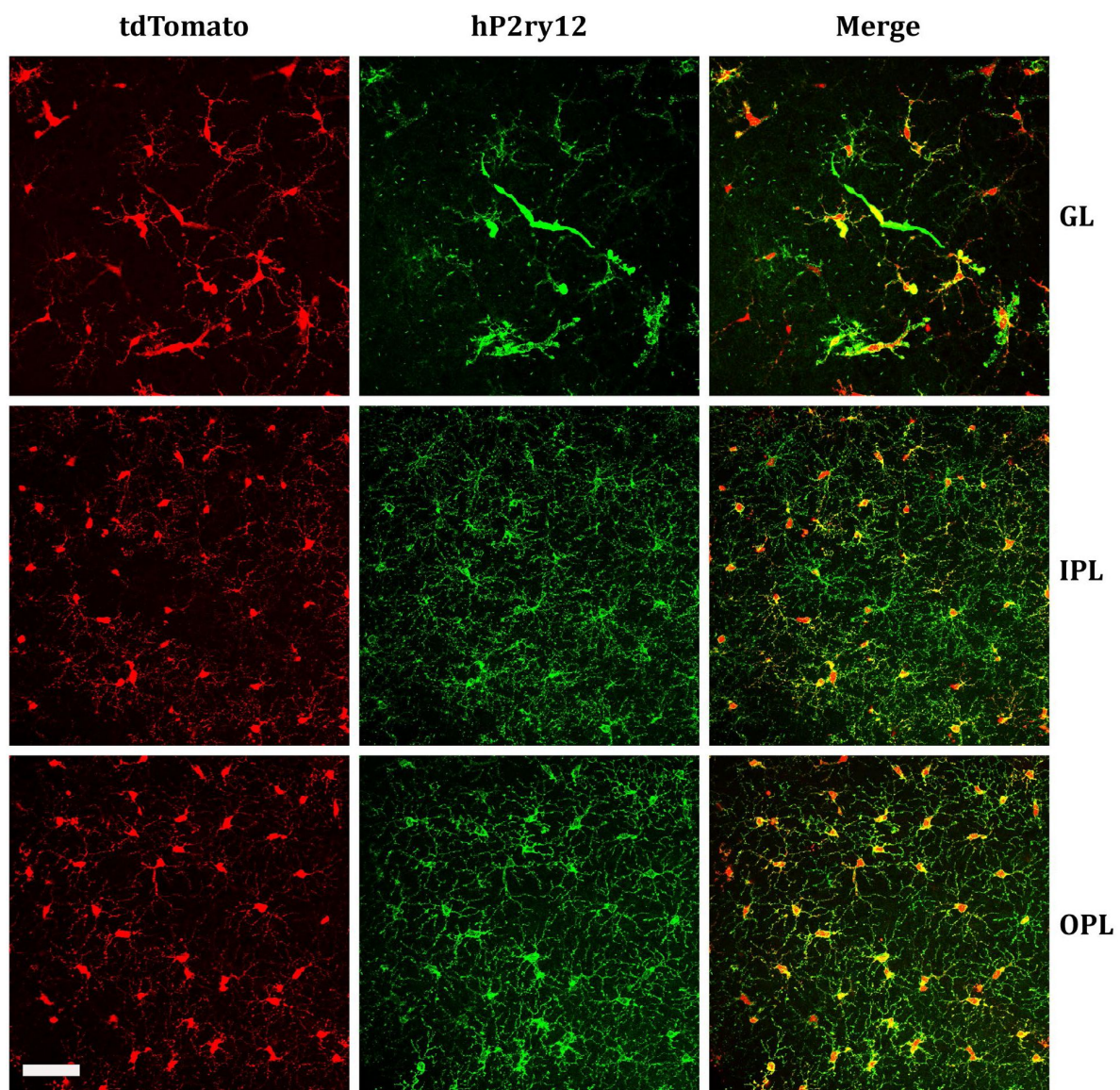


Fig. 6. Suppl A.

hP2ry12 staining on the retina after eight months of xenotransplantation. The images showed the tdtomato+ human microglia cells colocalized with hP2ry12 staining in GL, IPL, and OPL in the mouse retina. Scale bar = 300μm.

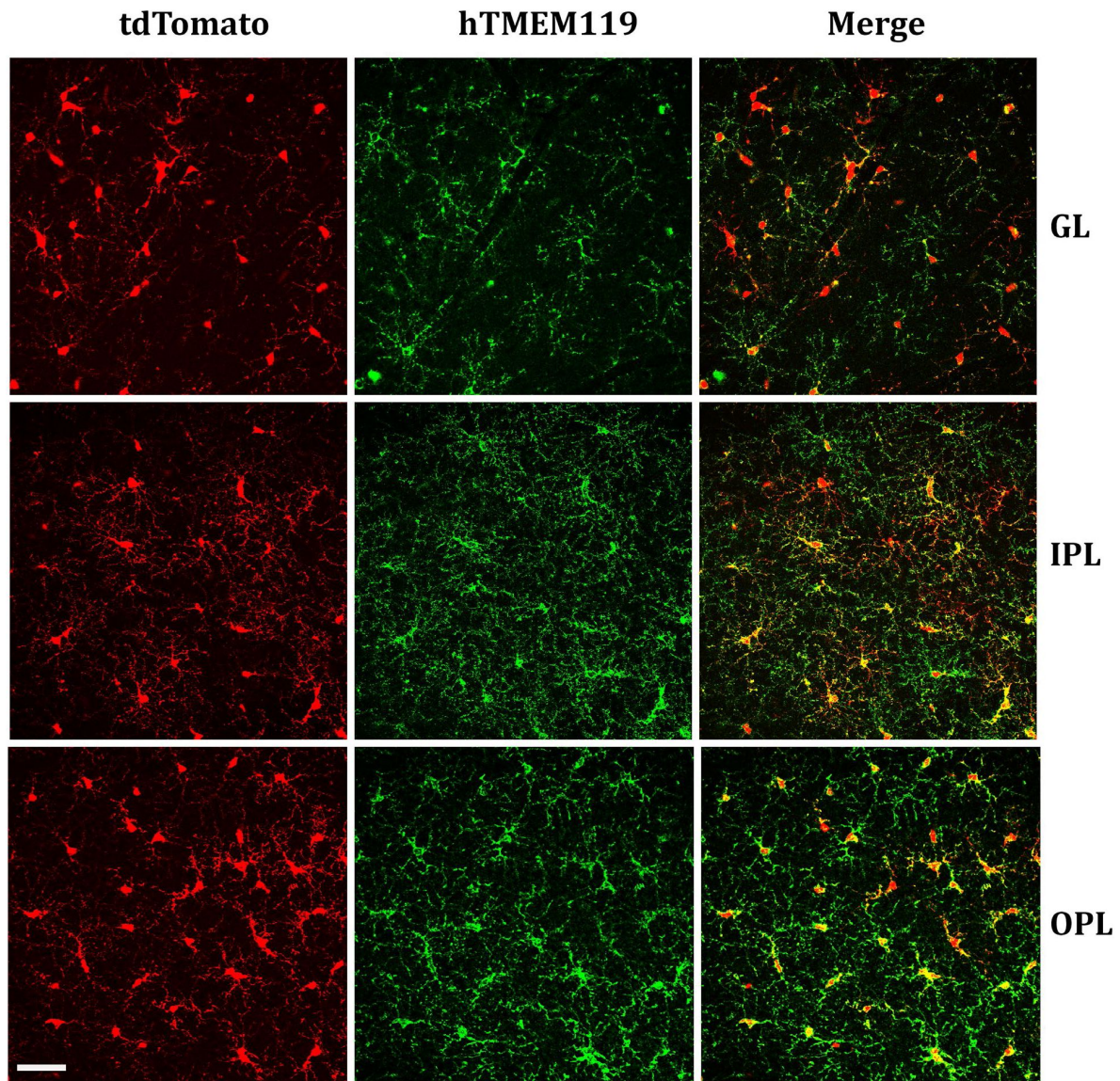


Fig. 6. Suppl B.

High magnification of hTMEM119 staining on the retina after eight months of xenotransplantation. The images showed the tdtomato+ human microglia cells colocalized with hTMEM119 staining in GL, IPL, and OPL in the mouse retina. Scale bar = 300µm.

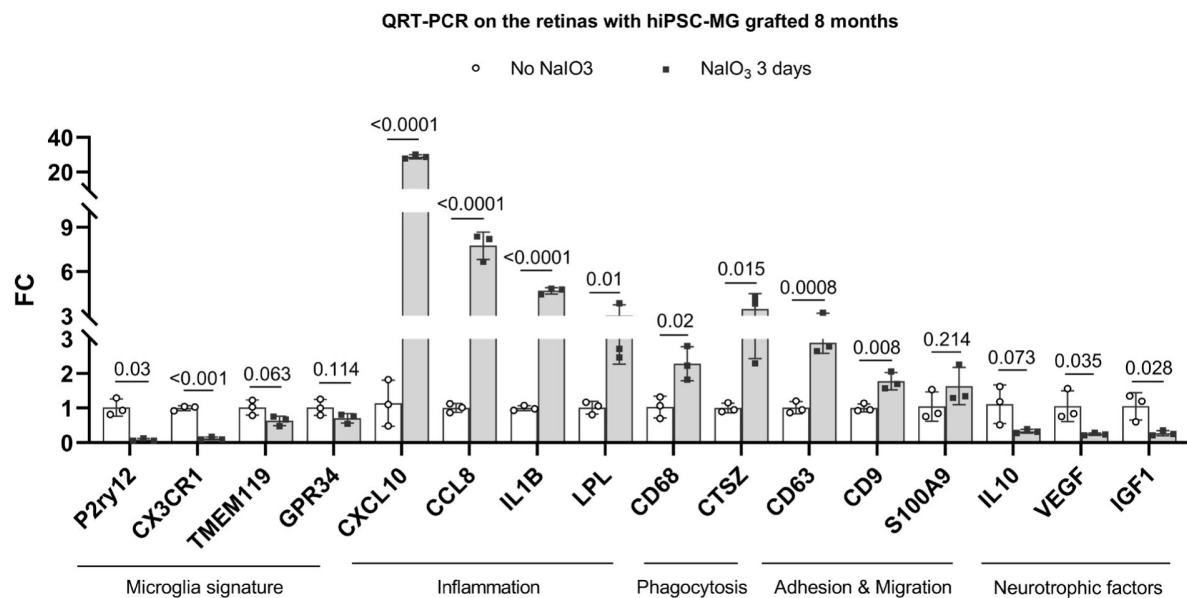
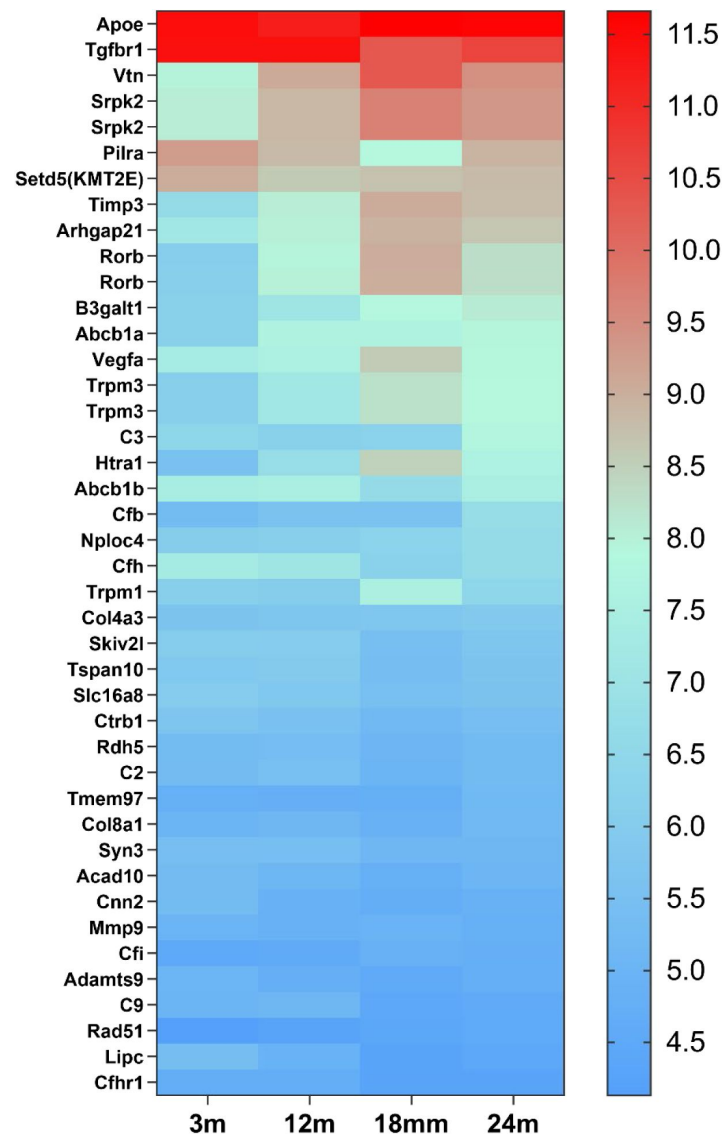


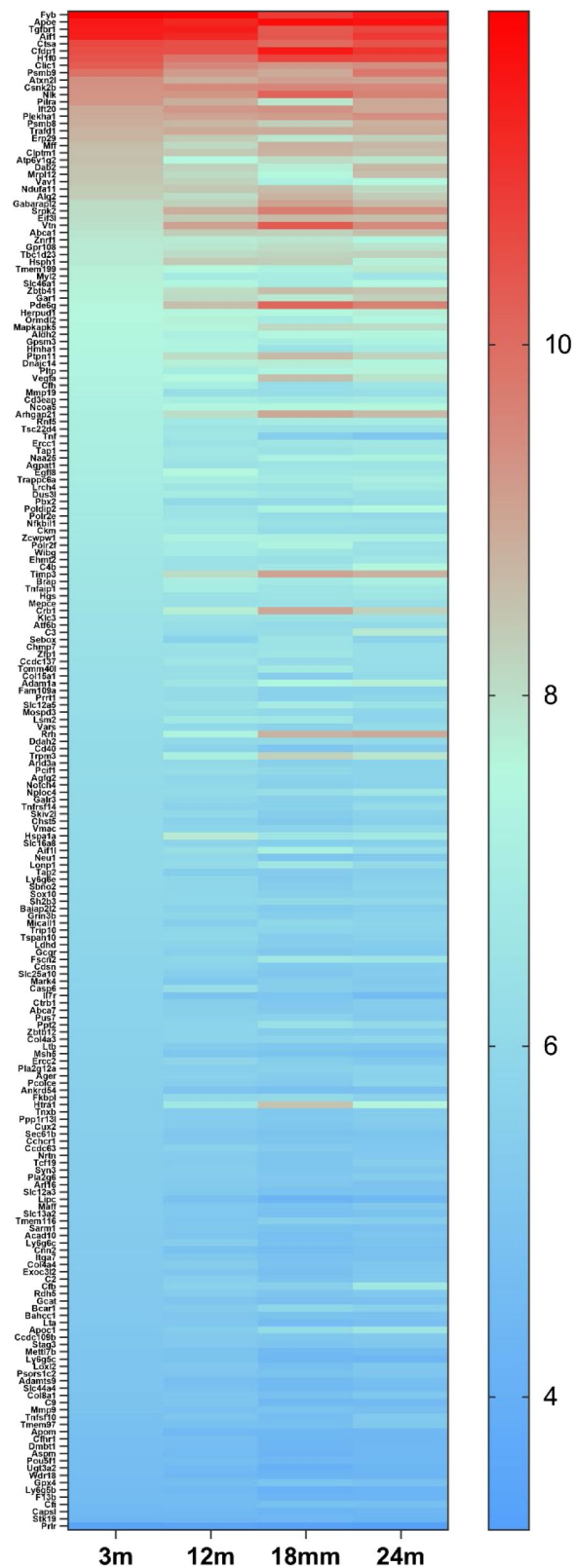
Figure 7. Suppl.

The inflammation, phagocytosis, adhesion and migration, neurotrophic factors, and microglia signature gene expression in hiPSC-derived microglia cells of grafted retinas. We compared gene coding sequences between humans and mice, chose a human-specific sequence to make the oligos (Suppl. Table 5), and ran a qRT-PCR on 8-month hiPSC-derived microglia cells grafted retinas with/without NaIO₃-treated retinas. The results revealed that hiPSC-derived microglia cells expressed more inflammatory factors and phagocytosis genes and promoted cell migration but decreased microglia cell signature genes and neurotrophic factors in NaIO₃-treated retina.



Discussion suppl A.

The heat map of 42 candidate genes from 34 loci associated with AMD expressed in retinal microglia cells. The microglia gene expression data are from microarray data previously published (Ma et al., 2013 [link](#)). The candidate genes came from the published paper (Den et al., 2022). The gene list is in Supplement Table 6.



Discussion suppl B.

The heat map of 209 genes associated with AMD (Fritsche et al., 2016) expressed in retinal microglia cells. The microglia gene expression data are from microarray data previously published (Ma et al., 2013). The gene list is in Supplement Table 7.

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Reviewer #1 (Public Review):

Summary:

This paper reported a protocol of using human-induced pluripotent stem cells to generate cells expressing microglia-enriched genes and responding to LPS by drastically upregulation of proinflammatory cytokines. Upon subretinal transplantation in mice, hiPSC-derived cells integrated into the host retina and maintained retinal homeostasis while they responded to RPE injury by migration, proliferation, and phagocytosis. The findings revealed the potential of using hiPSC-derived cell transplantation for microglia replacement as a therapeutic strategy for retinal diseases.

Strengths:

The paper demonstrates a method of consistently generating a significant quantity of hiPSC-derived microglia-like cells for in vitro study or for in vivo transplantation. RNAseq analysis offers an opportunity for comprehensive transcriptome profiling of the derived cells. It is impressive that following transplantation, these cells were well integrated into the retina, migrated to the corresponding layers, adopted microglia-like morphologies, and survived for a long term without generating apparent harm. The work has laid a foundation for future utilization of hiPSC-derived microglia in lab and clinical applications.

Weaknesses:

(1) The primary weakness of the paper concerns its conclusion of having generated "homogenous mature microglia", partly based on the RNAseq analysis. However, the comparison of gene profiles was carried out only between "hiPSC-derived mature microglia"

and the proliferating myeloid progenitors. While the transcriptome profiles revealed a trend of enrichment of microglia-like gene expression in "hiPSC-derived mature microglia" compared to proliferating myeloid progenitors, this is not sufficient to claim they are "mature microglia". It is important that one carries out a comparative analysis of the RNAseq data with those of primary human microglia, which may be done by leveraging the public database. To convincingly claim these cells are mature microglia, questions to be addressed include how similar the molecular signatures of these cells are compared with the fully differentiated primary microglia cell or if they remain progenitor-like or take on mosaic properties, and how they distinguish from macrophages.

(2) While the authors attempted to demonstrate the functional property of "hiPSC-derived mature microglia" in culture, they used LPS challenge, which is an inappropriate assay. This is because human microglia respond poorly to LPS alone but need to be activated by a combination of LPS with other factors, such as IFN γ . Their data that "hiPSC-derived mature microglia" showed robust responses to LPS indeed implicates that these cells do not behave like mature human microglia.

(3) The resolution of Figs. 4 - 6 is so low that even some of the text and labels are hardly readable. Based on the morphology shown in Fig. 4 and the statement in line 147, these hiPSC-derived "cells altered their morphology to a rounded shape within an hour of incubation and rapidly internalized the fluorescent-labeled particles". This is a peculiar response. Usually, microglia do not respond to fluorescent-labeled zymosan by turning into a rounded-shaped morphology within an hour when they internalize them. Such a behavior usually implicates weak phagocytotic capacity.

(4) Data presented in Fig. 5 are not very convincing to support that transplanted cells were immunopositive for "human CD11b (Fig.5C), as well as microglia signature markers P2ry12 and TMEM119 (Fig.5D)" (line 167). The resolution and magnification of Fig. 5D are too low to tell the colocalization of tdT and human microglial marker immunolabeling. In the flat-mount images (C, D), hCD11b immunolabeling is not visible in the GCL or barely visible in the IPL. This should be discussed.

(5) Microglia respond to injury by becoming active and losing their expression of the resting state microglial marker, such as P2ry12, which is used in Fig. 6 for the detection of migrated microglia. To confirm that these cells indeed respond to injury like native microglia, one should check for activated microglial markers and induction of pro-inflammatory cytokines in the sodium iodate-injury model.

<https://doi.org/10.7554/eLife.90695.2.sa2>

Reviewer #2 (Public Review):

Summary:

Ma et al. employed a myeloid progenitor/microglia differentiation protocol to produce human-induced pluripotent stem cell (hiPSC)-derived microglia in order to examine the potential of microglial cell replacement as a treatment for retinal disorders. They characterized the iPSC-derived microglia by gene expression and in vitro assay analysis. By evaluating xenografted microglia in the partly microglia-depleted retina, the function of the microglia was further assessed.

Overall, the study and the data are convincing, and xenografted microglia were also tested in a RPE injury paradigm.

<https://doi.org/10.7554/eLife.90695.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

(1) The primary weakness of the paper concerns its conclusion of having generated "homogenous mature microglia", partly based on the RNAseq analysis. However, the comparison of gene profiles was carried out only between "hiPSC-derived mature microglia" and the proliferating myeloid progenitors. While the transcriptome profiles revealed a trend of enrichment of microglia-like gene expression in "hiPSC-derived mature microglia" compared to proliferating myeloid progenitors, this is not sufficient to claim they are "mature microglia". It is important that one carries out a comparative analysis of the RNAseq data with those of primary human microglia, which may be done by leveraging the public database. To convincingly claim these cells are mature microglia, questions need to be addressed including how similar the molecular signatures of these cells are compared with the fully differentiated primary microglia cell or if they remain progenitor-like or take on mosaic properties, and how they distinguish from macrophages.

We greatly appreciate the insightful comments and suggestions from the reviewers, which were instrumental in enhancing our data analysis and organization. In response to the feedback, we have updated the terminology from "mature microglia" to simply "microglia" while clarifying in our text that these are fully differentiated microglia under single-type cell culture conditions.

Guided by the reviewer's advice, we incorporated RNA-seq data from human brain microglia studies conducted by Dr. Poon and Dr. Blurton-Jones' Lab (Abud et al., Neuron, 2017) and Dr. Huitinga's Lab (van der Poel et al., Nat Commun, 2019). We then conducted a comparative analysis of the gene expression profiles between our fully differentiated hiPSC-derived microglia and those from fetal/adult brain microglia (see Fig.2. Suppl. B, C and D; Suppl. table 1 and table 2). The correlation analysis revealed that our hiPSC-derived microglia closely resemble fetal and adult brain microglia, distinguishing them significantly from monocytes and inflammatory monocytes.

(2) While the authors attempted to demonstrate the functional property of "hiPSC-derived mature microglia" in culture, they used LPS challenge, which is an inappropriate assay. This is because human microglia respond poorly to LPS alone but need to be activated by a combination of LPS with other factors, such as IFN γ . Their data that "hiPSC-derived mature microglia" showed robust responses to LPS indeed implicates that these cells do not behave like mature human microglia.

We appreciate the feedback received. In response, we cultured hiPSC-derived microglia cells and subjected them to treatments with IFN γ , LPS, and a combination of both IFN γ +LPS, as illustrated in Figure 3 suppl. Our findings revealed that the IFN γ +LPS combination notably enhanced the expression of IL1a, IL1b, TNFa, CCL8, and CXCL10, whereas IL6 and CCL2 levels remained unchanged. Treatment with IFN γ alone significantly elevated the expression of TNFa, CCL8, CXCL10, and CCL2. These outcomes align with the findings reported by Rustenhoven et al. (Sci Rep, 2016), suggesting that the functionality of our hiPSC-derived microglia cells closely mirrors that of primary human adult microglia cells.

(3) The resolution of Figs. 4 - 6 is so low that even some of the text and labels are hardly readable. Based on the morphology shown in Fig. 4 and the statement in line 147, these hiPSC-derived "cells altered their morphology to a rounded shape within an hour of

incubation and rapidly internalized the fluorescent-labeled particles". This is a peculiar response. Usually, microglia do not respond to fluorescent-labeled zymosan by turning into a rounded shape within an hour when they internalize them. Such a behavior usually implicates weak phagocytotic capacity.

Thank you for your insightful comments. During submission, the main text's PDF version was converted online, resulting in low-quality output. We have since updated this with a high-resolution version. The observed alterations in cell morphology following zymosan phagocytosis may be attributed to the high zymosan concentration used (2mg/ml). We conducted an assessment to understand the impact of zymosan concentration on the morphology of hiPSC-derived microglial cells, as shown in Figure 4 suppl B. Our findings indicate that microglia cells adopt an amoeboid, rounded shape at zymosan concentrations exceeding 20ug/ml. To clarify this point, we have amended the text to read: "The cells altered their morphology and rapidly internalized the fluorescent-labeled particles."

(4) Data presented in Fig. 5 are not very convincing to support that transplanted cells were immunopositive for "human CD11b (Fig.5C), as well as microglia signature markers P2ry12 and TMEM119 (Fig.5D)" (line 167). The resolution and magnification of Fig. 5D is too low to tell the colocalization of tdT and human microglial marker immunolabeling. In the flat-mount images (C, I), hCD11b immunolabeling is not visible in the GCL or barely visible in the IPL. This should be discussed.

We are grateful for the reviewer's comments. As previously mentioned, the low quality of the images was due to the online conversion of the PDF version. We have now submitted both high-quality PDF and Word versions for the reviewer's assessment. In these high-quality versions, the colocalization of tdT with human P2ry12 and TMEM119 is distinctly visible. Additionally, we have updated the hTMEM119 staining images in Figure 5D. The results from hCD11b staining align with those observed in mouse CD11b staining, notably showing more effective staining in the outer plexiform layer (OPL) microglia cells. The reason for this—whether it pertains to a staining issue, a variance in CD11b expression among microglia cells in the OPL and ganglion layer (GL), or differences in the samples due to varying conditions—is not yet clear and warrants further investigation.

(5) Microglia respond to injury by becoming active and lose their expression of the resting state microglial marker, such as P2ry12, which is used in Fig. 6 for detection of migrated microglia. To confirm that these cells indeed respond to injury like native microglia, one should check for activated microglial markers and induction of pro-inflammatory cytokines in the sodium iodate-injury model.

The reviewer's insights are spot-on. We utilized preserved retinas to extract mRNA, which was then reverse-transcribed to cDNA for conducting qRT-PCR using human-specific primers, as detailed in the updated Table 5. The findings revealed that following retinal pigment epithelium (RPE) injury for 3 days, the transplanted hiPSC-derived microglial cells exhibited an increase in the production of inflammatory cytokines and upregulated genes related to phagocytosis, migration, and adhesion. Conversely, there was a decrease in the expression of microglia-specific signature genes and neurotrophic factors, as demonstrated in Figure 7 suppl.

Reviewer #1 (Recommendations For The Authors):

Line 52: "Microglia cell repopulation research suggests that: 1) if no injury or infection occurs, retinal microglia cells can sustain their homeostasis indefinitely" - this statement is too strong or delivers a confusing message; it needs clarification or to be backed up by evidence. Recent single cell RNA sequencing analyses suggest that even under a normal

condition, residential microglia do not present as a single homeostatic cell cluster, rather a subpopulation of activated inflammatory microglia are constantly detectable in the normal retina. This is likely because normal retinal neurons can be stressed due to various reasons, such as the temporal accumulation of misfolded proteins, exposed to strong light, or ageing, etc.

We appreciate the comments. We changed the sentence to read, "Microglia cell repopulation research suggests that: 1) retinal resident microglia cells can sustain their population with the local dividing and migration if any perturbations do not exceed the threshold of the recovery speed by local neighbor microglia cells."

Line 83: "we applied an appropriate protocol for culturing human iPSC-derived microglia cells" - it would be more appropriate if the word "appropriate" can be replaced by either "unique" or a phrase like "we adopted a (previously published) protocol..."

Thanks! We changed it to "We modified a previously published protocol to culture human iPSC-derived microglia cells."

Fig. 1F,G: A method of flow cytometry will provide more comprehensive cell quantification for percentages of positively labeled cells than cell counts under high magnification confocal images.

Thanks for the comments! We agreed with the reviewer. Given the experimental resources available, the quantifications of confocal images did provide a reasonable assessment. We will perform flow cytometry analysis in future experiments.

Reviewer #2 (Public review):

Weaknesses:

Gene expression analysis of mature microglia cells should be better interpreted and it would be beneficial to compare the iPSC-derived microglia gene set to a human microglial cell line (for example, HMC3) instead of myeloid progenitor cells. The way that the manuscript has been written, unfortunately, is not optimal. I recommend that the entire manuscript be edited and proofread in English. The text contains spelling and grammar mistakes, and the manuscript is inconsistent in several parts. The manuscript should also be revised for a scientific paper format.

We appreciate the reviewer's comments and have taken them into consideration along with similar inquiries from Reviewer 1. Following the suggestions, we conducted a comparison of gene expression profiles between our hiPSC-derived microglia and those from fetal/adult brain microglia, as depicted in the updated Fig.2. Suppl. B, C and D; as well as in the Suppl. table 1 and table 2. The correlation analysis demonstrated that the hiPSC-derived microglia cells closely resemble fetal and adult brain microglia, significantly differing from monocytes and inflammatory monocytes. Additionally, we have revised the manuscript to adhere more closely to the conventional scientific format.

Reviewer #2 (Recommendations For The Authors):

Specific suggestions for improvement:

- Regarding the characterization of human iPSC-derived microglia, P2RY12 is a general hematopoietic cell marker. One cannot judge the maturity of microglia only by P2RY12 expression (for example, line 261). The expression of more specific markers such as TMEM119 and PROS1 should be studied and discussed.

We are thankful for the reviewer's valuable feedback. In response:

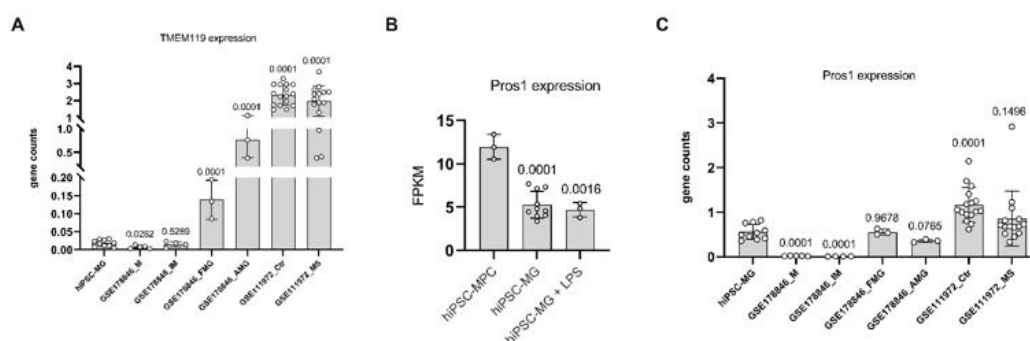
We have removed the term "mature" and clarified that the hiPSC-derived microglia we studied are fully differentiated within single-type cell culture conditions.

We performed a comparative analysis of the gene expression profiles between our hiPSC-derived microglia and microglia from human brains, as illustrated in the updated Fig.2. Suppl. B, C and D. The results affirm that hiPSC-derived microglia closely resemble human fetal and adult microglia.

We noted that the expression of TMEM119 in hiPSC-derived microglia under in vitro single-type cell culture conditions is notably low, as shown in the below A. This suggests that the stimulatory factors in our single-type cell culture might not sufficiently induce TMEM119 expression in microglia. The necessity for a retinal environment or interaction with neuronal and/or other glial cells for TMEM119 expression mirrors the behavior of infiltrating peripheral monocytes in pathological conditions, which initially lack TMEM119 but later differentiate into microglial-like macrophages that express TMEM119, as reported by Ma et al. in Sci Rep (2017).

Additionally, our findings suggest that PROS1 is not uniquely characteristic of microglia but is expressed across a variety of cell types. Within our specific culture conditions, we noted a higher expression of PROS1 in microglial progenitor cells, as shown in Author response image 1B and C.

Author response image 1.



- In Figure 2, Part E, the names of the genes or pathways in the figure are not clear, and are these genes the set that are the most differentially expressed between iPSCs-derived microglia and MPC? The analysis needs more explanation.

We regret any confusion caused by our previous explanation. To clarify, we compiled a list of microglia-enriched genes from the research conducted by Barres BA Lab (Bennett et al., Proc Natl Acad Sci U S A, 2016) and from our own RNA sequencing data of mouse retinal microglia, identifying a total of 130 genes predominantly expressed in microglia (Suppl. Table 3). We then applied this gene list to analyze our hiPSC-derived microglia RNA sequencing data, resulting in the identification of 71 microglia-specific genes. These 71 genes were subjected to Ingenuity Pathway Analysis (IPA) to visualize the signaling pathways involved. The details of these microglia genes can be found in the updated suppl. table 3.

- Lines 124 to 128 mention that high expression of Stat3, IL1b, and IL6 and their central role in pathway analysis emphasize the efficiency of the maturation protocol. Regarding the fact that Stat3, IL1b, and IL6 are contributors to proinflammatory pathways, it is not

convincing that the high expression of these genes in iPSC-derived microglia demonstrates the efficiency of the maturation protocol, given that microglia are not stimulated.

Thanks for the comments! We added the sentences about the comparison results between hiPSC-derived microglia and human brain microglia. We have also replaced the “mature” with “functional.” The sentence reads, “Thus, our method of obtaining differentiated microglia is a reliable method to generate a large number of homogenous functional microglia cells.”

- Statistical analysis is missing for some graphs, for example, figures 1-3 and 5.

We appreciate the comments. We have added the statistical results in the revised version.

- The legend for Figure 3 needs to be rewritten. The graphs or applied assays should be explained in the legend, not the interpretation of the data.

The legend was rewritten.

- There is no Figure 3 in the supplement figures file.

We added Figure 3. Suppl.

- hTMEM119 staining in Figure 5, Part D, is mostly background. Please provide another image.

The images were unclear after on-line converting due to the low number of pixels. We replaced them with new hTMEM119 staining images in Figure 5D.

- In line 176, figure 5I has been forgotten to be mentioned.

Thank you very much! We added 5I.

- Lines 241 to 244 state that more than 50% of the AMD-associated genes are highly expressed in retinal microglia according to Fig. discussion suppl A & B. It is not clear that the gene set that was used for analysis is from a healthy retinal microglia or AMD-related ones. Please explain precisely.

Thank you for your feedback. The gene list we referenced originates from a Genome-Wide Association Study (GWAS) that compared patients with Age-related Macular Degeneration (AMD) to healthy cohorts. We did not directly utilize this list in our experiments but referred to it to underscore the importance of microglia cells in the context of AMD.

Some of the English proofreading and manuscript format comments:

Line 805: Iba1 is written in lowercase. Is it human IBA1? It is not consistent with the way it is written in the text (in line 117, for example).

Thank you for pointing out the error. We reformed all Iba1 as “Iba1”. The Iba1 we used here are all from Wako (#019–19741), which labels both mouse and human microglial cells.

Line 814: microglia-enriched gene expression instead of microglia-enrich gene expression

Thank you! We changed it.

| *Line 345: Starting a sentence with lower case letter.*

Thank you! We changed it.

| *Line 342: Myeloid lineage instead of myeloid cell lineage.*

Thank you! We changed it.

| *Line 815: What does FPKM stand for? The abbreviations should be explained.*

The FPKM is the abbreviation of Fragments Per Kilobase of transcript per Million mapped reads. We added it in the text.

| *Line 309: The manuscript has occasionally referred to PLX-5622 without a minus. Please follow a uniform format.*

We changed all “PLX5622” to “PLX-5622”.

| *Lines 327-331: should be rewritten.*

The mentioned paragraph was rewritten.

| *Lines 335-340: should be rewritten.*

The mentioned sentence was rewritten.

| *Line 135: qRT-PCR instead of QPCR," as it is also mentioned in the methods and material. The correction also applies to all the QPCRs in the text.*

We changed “QPCR” with “qRT-PCR”

| *Figure 3: Graph B should be right side of graph A*

| *Images description: It is better to have the images description in the left side of the image, for example, figure 5 part B, GL, IPL and OPL*

Thanks for the suggestion. We changed the image organization as per the reviewer’s advice.

| *Lines 258 to 260 in the discussion have also been repeated with the same words in the introduction.*

The mentioned paragraph was rewritten.

| *Lines 327-331 should be rewritten.*

The mentioned paragraph was rewritten.

| *Lines 335-340 should be rewritten.*

The mentioned paragraph was rewritten.

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